



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



Automated Student Attendance Monitoring and Analytics System for Colleges

A PROJECT REPORT

Submitted by

Tarun S- 20221ISE0030

Gopi Krishna- 20221ISE0033

Abinaya K- 20221ISE0007

Under the guidance of,

Ms.JIJINA MT

BACHELOR OF TECHNOLOGY

IN

INFORMATION SCIENCE AND ENGINEERING

(AI AND ROBOTICS)

PRESIDENCY UNIVERSITY

BENGALURU

April 2026



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report “Automated Student Attendance Monitoring and Analytics System for Colleges” is a bonafide work of **Tarun S (20221ISE0030), Gopi Krishna (20221ISE0033), Abinaya K(20221ISE0007)** , who have successfully carried out the project work and submitted the report for partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in **INFORMATION SCIENCE AND ENGINEERING (AI AND ROBOTICS)** during 2025-26.

Ms. Jijina M T

Assistant Professor

Project Guide

PSCS

Presidency University

Ms. Sumita Guddin

Assistant Professor

Program Project Coordinator

PSCS

Presidency University

Dr. Gopal Krishna Shyam

Professor & Head of the Department

PSCS-COM,ISE.

Presidency University

Dr. Duraipandian N

Dean

PSCS

Presidency University

Sl. no	Name	Signature	Date
1			
2			

**PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND
ENGINEERING**

We the students of final year B.Tech in **INFORMATION SCIENCE AND ENGINEERING (AI AND ROBOTICS)** at Presidency University, Bengaluru, named **Tarun S (20221ISE0030), Gopi Krishna (20221ISE0033), Abinaya K(20221ISE0007)**, hereby declare that the project work titled “**Automated Student Attendance Monitoring and Analytics System for Colleges**” has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in **INFORMATION SCIENCE AND ENGINEERING (AI AND ROBOTICS)** during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Abinaya K 20221ISE0007

DATE: 24/04/2026

ACKNOWLEDGEMENT

For completing this project work, I have received support and guidance from many individuals, to whom I would like to express my deep sense of gratitude and indebtedness. I extend my sincere gratitude to our beloved **Chancellor, Vice Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in the completion of the project.

I express my cordial thanks to **Dr. Duraipandian N, Dean, Presidency School of Computer Science and Engineering**, and the Management of Presidency University for providing the required facilities and an intellectually stimulating environment that aided in the completion of my internship work.

I am also thankful to **Dr. Gopal Krishna Shyam, Professor and Head of the Department**, Presidency School of Computer Science and Engineering, Presidency University, for his mentorship and encouragement.

I would like to sincerely thank my internal guide **Ms. Jijina M T ,Assistant Professor**, Presidency School of Computer Science and Engineering, Presidency University, for the moral support, motivation, timely guidance, and encouragement provided during the period of project work.

I am grateful to **Ms. Sumita Guddin, Program Project Coordinator**, Presidency School of Computer Science and Engineering, for facilitating problem statements, coordinating reviews, monitoring progress, and providing valuable support and guidance.

I also extend my sincere thanks to the **Teaching and Non-Teaching staff** of the Presidency School of Computer Science and Engineering, and staff from other departments, for their valuable help and cooperation.

Tarun S

Gopi Krishna

Abinaya K

Abstract

Traditional methods have been used to record attendance in most learning institutions either using paperwork (manual registers) or even basic computerized entries which may be hectic and also easily erroneous. The proposed project will introduce an Automated Student Attendance Monitoring and Analytics System, which will be used to streamline and enhance the attendance process with face recognition and RFID authentication. The system is aimed at saving man power, eliminating proxy attendance and giving accurate and reliable record of attendance.

This system that is proposed involves the use of a camera that takes pictures of students in a classroom and the use of face recognition methods to identify the students. Besides this RFID cards are also an added verification measure to ensure that the attendance taken is authentic. Upon identification of the student, the system clocks the attendance and the system records the data in a database and date and time.

There is also a component of analytics included in the system and is useful in monitoring attendance trends and students with low attendance. This data may be helpful to faculty and administrators to track student attendance and implement adequate measures at an initial stage. Attendance records are also available to students and they are therefore informed about their academic needs.

This system implementation demonstrates that attendance may be recorded in more efficient and systematized way as compared to the traditional means. It saves on time taken in taking attendance in lecture halls and enhances accuracy through minimization of human errors. Overall, the system offers a viable way of streamlining the process of attending management in institutions of learning.

Besides the existing functionality, the system can be expanded and extended to a broader usage. It can also be upgraded to accommodate cloud based storage so that the attendance data can be accessed elsewhere. The system can be made easier to use by students and faculty by features like real time attendance alerts when it becomes low and allowing access to the mobile application. The other academic modules can also be coupled with the system in order to form a more complete academic management system. These enhancements allow the proposed system to apply in the actual world learning settings and aid in the improved monitoring and decision making.

Table of Content

Sl. No.	Title	Page No.
	Declaration	I
	Acknowledgement	II
	Abstract	III
	List of Figures	VI
	List of Tables	VII-VIII
	Abbreviations	IX
1.	Introduction	1
	1.1 Background	1
	1.2 Statistics of project	1-2
	1.3 Prior existing technologies	2
	1.4 Proposed approach	2-3
	1.5 Objectives	3-4
	1.6 SDGs	4
	1.7 Overview of project report	5
2.	Literature review	6-15
3.	Methodology	16-20
4.	Project management	21
	4.1 Project timeline	21
	4.2 Risk analysis	22-25
	4.3 Project budget	25-26
5.	Analysis and Design	27
	5.1 Requirements	27
	5.2 Block Diagram	29
	5.3 System Flow Chart	30-31
	5.4 Choosing devices	32-33
	5.5 Designing units	33-35
	5.6 Standards	36-37
	5.7 Mapping with IoTWF reference model layers	37-39
	5.8 Domain model specification	39-40
	5.9 Communication model	40-41
	5.10 IoT deployment level	41-42

	5.11 Mapping IoT deployment level with functional view	42-44
	5.12 Operational view	44-45
	5.13 Other Design	45-46
6.	Hardware, Software and Simulation	47
	6.1 Hardware	47-48
	6.2 Software development tools	48-49
	6.3 Software code	49-50
	6.4 Simulation	50-52
7.	Evaluation and Results	53
	7.1 Test points	53-54
	7.2 Test plan	54
	7.3 Test result	54-55
	7.4 Insights	55
8.	Social, Legal, Ethical, Sustainability and Safety Aspects	56
	8.1 Social aspects	56
	8.2 Legal aspects	56-57
	8.3 Ethical aspects	57
	8.4 Sustainability aspects	57
	8.5 Safety aspects	58
9.	Conclusion	59
	References	60-61
	Appendix	62-64

List of Figure

Figure	Caption	Page no
1.1	Sustainable development goals	5
3.1	The V model methodology	19
3.2	Another example of the V model methodology	20
4.1	Project Timeline (Gantt Chart Representation)	21
4.2	PESTLE analysis	22
4.3	Risk distribution by category	25
5.1	Functional block diagram	30
5.2	System Flow Chart of the Proposed Attendance System	31
5.3	Functional Unit Interaction in the Attendance System	35
5.4	IoT World Forum Reference Model Layers	38
5.5	Communication Model of the Attendance System	41
5.6	Operational View of the Attendance System	45
6.1	Simulation Output of Attendance System	52

List of Tables

Table no	Caption	Page no
Table 2.1	Summary of literature reviews	14-15
Table 3.1	Mapping of project stages to the V-Model	17-18
Table 4.1	Project implementation timeline	22
Table 4.2	PESTLE analysis	23
Table 4.3	Risk matrix	23-24
Table 4.4	Estimated project budget	25-26
Table 5.1	Summary of System Requirements	28-29
Table 5.2	Comparison of Microcontrollers	32
Table 5.3	Camera Comparison	33
Table 5.4	RFID Module Comparison	33
Table 5.5	Standards Used in the System	36-37
Table 5.6	Mapping Project Layers with IoTWF Reference Model	38-39
Table 5.7	Description of Domain Model	40
Table 5.8	Communication Model Summary	41
Table 5.9	IoT Deployment Components	42
Table 5.10	Mapping of Deployment Level and Functional Groups	43-44
Table 5.11	Additional Design Components	46
Table 6.1	Hardware Components Used	47-48
Table 6.2	Software Development Tools	49
Table 6.3	Simulation Test Results	51
Table 7.1	Test Points	53

Table 7.2	Test Results	55
-----------	--------------	----

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
RFID	Radio Frequency Identification
IoT	Internet of Things
IoTWF	Internet of things World Forum
SDG	Sustainable Development Goal
LAN	Local Area Network
USB	Universal Serial Bus
CPU	Central Processing Unit
DBMS	Database Management System
UI	User Interface
API	Application Programming Interface
CNN	Convolution Neural Network
LBP	Local Binary Patterns
GPS	Global Positioning System
QR	Quick Response
IDE	Integrated Development Environment

Chapter 1

Introduction

1.1 Background

The contemporary educational environment has incorporated the use of technology with an aim of transforming modes of knowledge delivery and consumption. However, one of the areas of concern that have remained stuck to the archaic manual methods is the student attendance management. The traditional method of attendance recording still relies much on the services of physical registers or verbal roll calls in most institutions of higher learning particularly the large scale colleges. This is not a routine matter as far as administration is concerned but a segment of the larger goal of fostering academic discipline and student success. In making a faculty member spend considerable amount of his or her teaching time on taping presence, one will create a bottleneck that will occur repeatedly, which will have an adverse effect on the quality of the learning experience. The suggested way to fill in the specified gap is the Automated Student Attendance Monitoring and Analytics System that will transform an otherwise boring process into a complicated data-driven procedure that will help with the eventual goal of the modern education. With the computerization of this aspect of campus life, institutions will be capable of ceasing reactive monitoring and would decide to take a proactive stance where attendance records would be an important instrument in gauging student interest and wellness of the institution.

1.2 Statistics

The quantitative study of the efficiency of the institutional institutions has shown the immense invisible costs of manual attendance tracking systems. Indicatively, one of the most common observations in different studies is that the faculty members normally lose some ten minutes out of a sixty minutes lecture in calling out names or distributing signature sheets. Such wastage of time, when summed up in a given academic department during one semester, is a huge wastage of hundreds of instruction hours and educational resources. Also, human error often jeopardizes the accuracy of manual records when transcribing information on paper logs into digital databases, and some institutional audits have indicated an error rate of up to fifty-two percent. The relationship between the academic achievement and the attendance of students is also statistically and measurably directly related. Statistics always prove this to be true as students that attend school with a good attendance rate (more than eighty five percent attendance) achieve much higher scores on tests than

those with less attendance. In addition to performance indicators, institutional data is misconstrued because of the problem of proxy attendance, and the management is unable to do the right academic counseling or discern students that might be at risk of dropping out. The automation of these statistical anomalies can help an institution to enhance the integrity of its records and create a more disciplined atmosphere in the student body.

1.3 Prior existing technologies

Until the development of specialized automated solutions, learning institutions mostly used physical attendance registers. This is the customary approach where a faculty member needs to go and call names out by hand or go round with a sheet of paper and signatures. Although this method is very easy, it is fundamentally flawed as it eats almost fifteen percent of each lecture and is extremely prone to buddy punching, in which a student registers an absent classmate. The first change was to semi-digital systems like Microsoft Excel spread sheets or Google Forms even as technology entered the classroom. Although the tools were beneficial towards the digitalization of the data, they did not address the primary issue of manually entering data. A teacher still had to enter information on paper into the computer which in most cases caused huge errors in data entry and a huge administrative burden at the end of every semester. Some of the institutions in recent years tested hardware-based frameworks, such as Radio Frequency Identification (RFID) cards. An RFID system is the one whereby the students swipe their ID cards on a scanner when entering the room. Although this is quicker than a roll call, it is not effective in the prevention of proxy attendance as any one student can easily carry and scan a number of cards on behalf of his or her peers. On the same note, first biometric systems that used fingerprint scanners were launched, although they were prone to hygienic issues and mechanical malfunctions because the number of daily transactions was very high. The current technologies are mostly dedicated to the process of capturing the presence and do not offer any meaningful results or some real-time notifications to the management, leaving gaping holes in monitoring and academic counseling of the students.

1.4 Proposed approach

The suggested system will overcome the weaknesses of the existing approaches because it proposes an integrated centralized platform that consolidates automated monitoring and advanced data analytics. This system will be much more than a digital replica of a paper register: it is meant to be a complete student engagement ecosystem. Its strategy is focused on a web-based dashboard, which joins faculty, students, and administrators in real-time. Automation of the data collection process

eliminates the human error that is present in the manual entry of the data and gives the system a benefit of finalizing the attendance records as soon as a classroom is in session. This near real-time synchronization will enable the institution to take action on the information when it remains relevant (e.g., an automatic alert to parents or advisors when a student has missed a specific percentage of classes).

In addition to mere tracking, the analytical engine is the main strength of this proposed solution. The system does not simply store numbers, but it processes them to determine patterns in the student behavior. An example of this is that using the analytics module, the department can identify particular days of the week or particular subjects that have a low attendance rate thus be able to investigate whether there may be a problem with the schedule or delivery of the curriculum. The system can convert raw data to actionable intelligence by giving the faculty visual reports and trend lines. It is this change of focus of the project towards recording, to that of analyzing that makes this project stand out as it directly helps the academic objective of increasing student retention and performance by proactively managing it in advance instead of reactively punishing it after the fact.

1.5 Objectives

The main vision of the project is the possibility to make an innovative breakthrough in the management and interpretation of student presence in an educational institution via a completely integrated digital ecosystem. The system on its fundamental level is aimed at eradicating the unnecessary administrative workload imposed on the faculty members at present and automating the whole process of attendance recording. The project will help to save precious teaching time that can be utilized to teach and interact with the students by removing this task, which was formerly handled manually, through a computer-based roll call program. One of the important secondary goals is complete elimination of proxy attendance which is still a major challenge in traditional classes. The system also provides the assurance of honest representation of classroom participation through secure authentication that ensures the data recorded remains a true reflection of classroom participation thus maintaining the academic integrity of the institution.

Moreover, the project will help create a bridge in communication between students, faculty and administration by availing a platform through which real time monitoring can be done. All the stakeholders will be provided with live data rather than waiting to the end of a month or a semester to check the attendance records. This openness is essential in the creation of holistic attendance analytics which is a potent academic decision-making instrument. The system offers faculty members

with the information necessary to interfere with the students and help them before their grades are adversely affected by trends and patterns of absenteeism. The end-game is to establish a high performing and disciplined academic atmosphere where attendance keeping is not an administrative practice but a strategic resource towards advancing the overall student achievement.

1.6 SDGS

The design of Automated Student Attendance Monitoring and Analytics System is not only a technical process but a move towards achieving a number of global sustainability goals. Sustainable Development Goal 4 is the goal that best fits the project since it focuses on the provision of inclusive and equitable quality education and the development of lifelong learning opportunities to everyone. The system aids learning institutions to detect under engaged or at-risk-of-failing students in the first semester by automating the tracking process and offering real-time analytics to the educational institution. Such preventative measure will make sure every student receives the support that they require in order to achieve success, thus increasing the retention rates and the standard of the output of the provided education.

The project, in addition to the classroom, leads to the Sustainable Development Goal 9 through digital infrastructure innovation and updating of the digital infrastructure of educational institutions. Enhancing campus campus operations that use advanced monitoring devices is an indication of a shift towards smarter campus environments that use data to govern. Moreover, the shift to the fully digital attendance registration system with traditional paper-based registers can help achieve Sustainable Development Goal 12 as it is aimed at responsible production and consumption. The system lowers the need to use paper and physical storage by a great margin, which is a more eco-friendly administrative model. The correspondence of these project objectives to the overall global system is graphically shown in Figure 1.1.



Fig 1.1 Sustainable development goals

1.7 Overview of project report

Chapter 1 introduces the motivation, background, and scope of the project.

Chapter 2 presents an extensive literature review covering prior works in speech disorder intervention and related technologies.

Chapter 3 discusses the methodology, including the development framework and techniques used for analysis.

Chapter 4 elaborates on project management, including risk analysis, budget, and timeline.

Chapter 5 covers system analysis and design, detailing the functional architecture and models.

Chapter 6 explains the software development tools, implementation process, and simulations performed.

Chapter 7 presents evaluation results and analysis of system performance.

Chapter 8 focuses on social, ethical, legal, and sustainability aspects of the project.

Chapter 9 concludes with the outcomes, key findings, and future work recommendations

Chapter 2

Literature review

The literature review will be used as the scholarly background of this project, which will provide an in-depth explanation of the history of the evolution of the current state of attendance management technologies. The analysis of a wide variety of studies, including the early examples of biometrics and more recent artificial intelligence systems, will allow defining the exact technical issues that have not been addressed yet. In this chapter, the researcher examines the current methodologies in order to establish how various researchers have tackled the issues of manual error, proxy attendance, and data transparency. In the end, this review enables us to situate our proposed system in the existing technological environment in that our solution will be based on previous successful practices with particular consideration of the gaps in past studies.

2.1 Real-Time Student Face Recognition Attendance System using AI [1]

The study carried out in the 2025 article named "Real-Time Student Face Recognition Attendance System using AI"[1] investigates the application of computer vision in the roll-call system in the classroom of the present days. The authors outline a framework that makes use of deep learning models to identify and record students who enter the lecture hall or at the session. The main benefit of this method, as explained in the paper, is that manual interference is removed and this in any case contributes a lot of time wastage in any academic hour. The system offers a high accuracy rate by capturing the facial features and matching them to a pre-registered database thus it minimizes the chances of proxy attendance as is the case with the traditional paper or card-based systems. The paper highlights the significance of the real-time processing in order to make sure that the administrative documentation is updated immediately without any postponements in commencing the lecture.

Even with the creative use of artificial intelligence, it is possible to see that some gaps can be found in this study. The most obvious one is the performance of the system in the non-ideal environment, like in the classroom where the lighting is not good, or where the students are not facing the camera lens. The paper observes that the recognition rates are reduced considerably in case of partial occlusion or when the students are facing downwards to get a view of their desks. To enhance this, the future versions may make use of more powerful facial landmark detection methods that are not sensitive to different angles and low-light

conditions. Moreover, the article fails to critically examine the ethical aspect of constant facial tracking or the security measures that would be in place to ensure that biometric information is not stolen. This project addresses these issues by focusing on a more balanced combination of software and hardware.

2.2 Automated Attendance and Analytics System [2]

The article by researchers in "Automated Attendance and Analytics System"[2] published in IJRASET concentrates on the integration of automated data collection and the further utilization of the information on the institutional intelligence. This paper does not consider attendance as a mere binary list of who has been present and absent, but states that the data can be analyzed to give information on behavior and engagement of students. The authors discuss a system that creates automatic reports to the faculty members in order to monitor the long term attendance pattern on both individual and class level. This processing of raw data into a consumable information is something that is offered as a key move towards an educational management of the modern era so that there can be early intervention among the students who might be in danger of failing academically due to chronic absenteeism.

Nevertheless, there is critical analysis of this system showing that it relies on semi-automated points of entry of data, and even then, the system remains subject to human error or manipulation at the point of entry. Although the analytical dashboard provides a holistic approach, the quality of results is limited to the quality of input data. There is no specific discussion on how to manage the real world classroom complexities like entering of students late and leaving early, which may influence the results of the analytical process to the side. The best way in which this can be enhanced in this methodology is by ensuring that a more convenient fully automated tracking system that will not need any manual input at all by the students or the staff is implemented. This would make the resulting analytics much more meaningful to the academic counseling and administrative decision making because of the full objectivity and automatization of the data source used. The proposed system is based on this idea as it develops a direct connection between high-integrity automated capture and a more refined analytical engine.

2.3 Automated Attendance Management Systems: Systematic Literature Review [3]

In the systematized literature review performed by Ali et al. [3], the authors give a graduate academic summary of the different technologies that have been deployed in automating the attendance tracking within the last ten years. The paper is of a special importance as it

classifies the existing systems into specific technological generations which span between simple barcode and RFID systems to a more modern biometric-based and mobile-based applications. The performance of these systems according to the cost, ease of deployment, and security are some of the criteria that the authors examine. They find that although the use of hardware intensive solutions such as fingerprint scanners offers a high security level, it may not be easily scaled in large university settings because of the cost of maintenance and hygiene issues. Conversely, GPS or QR based systems via mobile platforms have been detected to be more versatile but very vulnerable to spoofing and unauthorized alteration of data. This is because of the systematic character of this review and thus, it can be seen that the academic community is shifting to more integrated software-driven solutions that have less physical contact with hardware.

Nonetheless, one of the most critical gaps that were revealed in this entire review is that there is no research conducted with a longitudinal analysis of attendance data to predict behavior. Although the authors take a lot of time to explain how different systems capture the presence of a student, they have very minimal information on how institutions should use the data after it has been obtained. The majority of the systems reviewed are meant to be passive recorders and not active analysis systems. To build on the preliminary information presented in this research, the future study should be conducted on the creation of all-inclusive analytics modules that could correlate attendance patterns with other educational factors such as grade point averages or internal assessment scores. With the refocusing that is placed on the analysis rather than the act of recording, institutions will be able to turn basic sheets of attendance records into an effective instrument of diagnosing academic success. This gap is directly covered by our project that will develop a centralized analytics engine that will process raw attendance information to offer actionable insights to faculty and student counseling services.

2.4 Automated Student Monitoring and Attendance System [4]

The study outlined in Ghadge et al. [4] explores the use of a local student monitoring system that is intended to combine the attendance tracking and real time classroom management. The approach that this research paper outlines is a centralized software interface in which the faculty members would be able to track the attendance of certain batches in real-time. The main concept that is advanced in this piece is the focus on monitoring as the independent yet related role to the recording. The authors propose that a system must not only indicate an absent teacher but should also present the view of the whole level of engagement in the

classroom. Their study proves that the level of digital transparency incrementally improved the level of student punctuality and attendance rates, since the knowledge of a high-integrity monitoring system serves as a psychological deterrent against absenteeism.

Although the operation would be advantageous as debated in this paper, it has a number of inconsistencies and technical limitations that restrict its performance in a big institution. According to Ghadge et al., its described system is highly dependent on the local area network (LAN) architecture that causes serious challenges in terms of accessibility to management and parents who might require seeing the reports even outside of the station. Besides, the paper lacks automatic system of informing the external parties, including parents or guardians, of a student with a low attendance below the required threshold. This does not require any automated communication so that the administrative action is still manual and slow. One more update to this solution would be transferring the system to a cloud system, which will enable cross-platform synchronization and push notifications. The system would be a far more proactive instrument of institutional governance with the help of automated SMS or email notifications based on particular attendance quotas. These findings are the ones that our proposed work is aimed at expanding by making sure that our analytics dashboard is made as global as possible and that it is packed with the notification protocols that are automated.

2.5 An Automatic Student Attendance Monitoring System using Face Recognition [5]

The study offered by Osman et al. [5] in IET Smart Systems is a major breakthrough in the use of facial recognition in academic administrative works. The study approach presented in the 2024 paper is based on advanced image processing algorithms to consider students in a classroom environment. Their strategy is based on the core of a convolutional neural network, which has been trained in order to be able to differentiate between individual students within the context of a busy classroom setting. The high resolution cameras will be installed at fixed spots within the room which will be used by the system to scan the seating area at specific times during a lecture. This technique helps in eliminating the faculty interaction and as such the teaching process proceeds without interruption and the background system communicates with the digital register. The main point that the authors make is that this automation saves time, and it can also give a high rate of security against the proxying in the attendance because the biometric identification itself is unique to an individual.

Nevertheless, although the presented facial recognition models have high technical standards, the research demonstrates a few gaps in the practical implementation that should be filled. A key issue is the computational cost of processing multiple high-resolution video feeds in real-time which may not be affordable to institutions with small local server capability. The paper also pays much attention to the recognition stage but fails to give a comprehensive data management plan on long-lasting storage and retrieval. It is also interesting to note that the user interface design is significantly missing to enable students to check on their own attendance status in real time. In order to enhance this model, more efficient edge computing solution may be considered as the way of processing the images locally, and then transmitting only the metadata to the cloud. Moreover, mobile-responsive student dashboard would be a pleasant addition to the described system that would provide an additional layer of transparency. The improvements that our proposed project integrates include: lightweight data processing and offering a multi-user dashboard in order to engage better with the stakeholders.

2.6 Face Recognition Based Attendance System [6]

The article by Abraham et al. [6] of 2020 discusses a simpler form of the facial recognition technology, which can become affordable to smaller institutions of learning. The scientists used Haar Cascade classifier and Local Binary Patterns Histograms algorithm to carry out face detection and recognition. The process includes a mere arrangement where the images of students entering the classroom are caught by the use of a single camera. The system is designed to be user-friendly and very lightweight and therefore does not need the user to purchase expensive specialized hardware but can fit into normal personal computers. The major advantage of this work is its simplicity and the demonstration of the possibility to implement the automated attendance without spending a lot of money. The authors show that the number of manual errors can be reduced greatly when compared to the traditional paper based approaches using simple hardware.

However, the shortcoming of this method is apparent when the system is extended to bigger and more dynamic places. Since Haar Cascade algorithm is sensitized to alterations in facial expression, light and head position, the recognition accuracy varies in practical situations. There is also a major gap that the research finds in the system in dealing with the occlusions which include students wearing glasses or masks which are prevalent in campus setting. Moreover, the database management that is presented in the paper is somewhat unsophisticated and lacks in terms of more sophisticated analytics and automated reporting

capabilities. The next way that this research can be improved is to substitute the old classification algorithms with the more modern deep learning models that are more resistant to environmental fluctuations. Moreover, the system may be improved with the introduction of a powerful back-end database which would allow making sophisticated queries and trend analysis. In our work we aim to extend the affordability of such approach but improve the underlying technology and make it more reliable and better in analytical output.

2.7 Face Recognition Based Attendance Management System [7]

In the research paper that was written by Patel and Shah [7], the author has mentioned how facial recognition can be integrated into a wider management system and not just as an identification process. The process is multi stage and the methodology placing the face first by the Viola-Jones algorithm and then applying a database comparison to recognize the face. Among the brightest aspects of this work is the focus on the administrative background, in which the authors created a system that automatically feeds an Excel-based database. This linkage between the vision system and the data storage system is supposed to improve the administrative lag which frequently succeeds the manual attendance taking. The study reveals a high decrease of the labor hours spent in consolidating end of semester attendance by a digital connection between the classroom camera and the department records.

Although this management system is working efficiently, on a critical analysis, there is a loophole in the liveness detection ability of the software. The paper fails to discuss in details the methods of thwarting students who may use a high-quality photograph or a digital monitor to defraud the camera, which is a common weakness in typical face recognition systems. Also, although the system will refresh an existing database, it does not have a real time notification interface where the students will be provided with instant feedback about their attendance status. To enhance this piece of work, it can be supplemented by the incorporation of 3D depth sensors or infrared analysis of the subject to make sure that it is a tangible human being and not digitally reproduced. Moreover, it would be better to incorporate a web interface that will deliver automatic email summaries to the students at the conclusion of every week, which will contribute to the transparency of the system. Our project will attempt to address these drawbacks through the use of more contemporary spoofing defenses and a direct to user reporting dashboard.

2.8 Smart Attendance Management System Using Face Recognition [8]

In the article called Smart Attendance Management System Using Face Recognition by Kumar and Singh [8], the researcher offers a clever solution to the problem of the large

numbers of students that may appear during the busiest periods and the authors suggest to use high-speed recognition. The strategy involves optimization of the recognition algorithms to fit the low power computing devices with a view of rendering the technology feasible in the classroom settings where the high end servers are not available. The authors are also able to use the Open Source Computer Vision library in coming up with a system that would be able to process frames at the necessary speed to identify students as they go past a fixed point. The research notes that the smart system must not be noticeable to the user that is, students should not be requested to stop or stand in front of the camera but natural movement into the lecture hall is to be maintained.

Nevertheless, the aim to get the high speed processing in this study results to an evident gap in accuracy of recognition when many students are in the frame simultaneously. The paper also recognizes that crowd induced-indicates are another major challenge in which students who are passing by others are frequently overlooked by the system. Moreover, the system has been developed as an independent tool and does not provide any cross platform analytics so that the higher level management can make comparisons between the attendances of various departments or semesters. One possible way to enhance this study is to use a tracking algorithm that will be able to trace the movement of a student across several frames so that they can be recorded even when their face is temporarily covered. Also, it would enable the use of a cloud based analytics module to allow the use of the data not only as a daily record. Our project is based on the smart and invisible philosophy of this paper with an addition of a solid analytical layer that operates throughout the entire institution.

2.9 Automatic Attendance System using Face Detection and Recognition [9]

The study by Khan and Ahmad [9] is dedicated to technical peculiarities of the detection-recognition pipeline, namely to the problem of speed accuracy trade off of automated systems. The process of identifying a face in a frame and matching it with the database follows a two stage procedure in which the system is first required to identify the presence of a face in one frame and then a second algorithm is used to match it to the database. The paper is also unique in the way it examines various lighting scenarios and what can be done to design a system that is operable during the different light Moods of a normal classroom in a day. The authors test the software in real world environments and not in the controlled laboratory environments only to give a more realistic account of the performance of computer vision under the shadow of particular surroundings or under window glare. Their findings show that

the detection process is relatively constant but recognition process demands the high-quality image data in order to keep the false rejection relatively low.

Another major research gap that has been determined in the work, however, is the absence of the history aware feature in the tracking process. The system considers each frame as an instance; it thus tends to vary at times with the recognition of a student not being identified and vice versa within a short span of a second. To address this, a temporal filtering mechanism might be implemented to average the results of recognition across a number of seconds resulting in a more reliable and consistent log. Also, in the paper there is a strong emphasis on technical aspect of implementing the vision system which overlooks the creation of an administrative interface in order to generate reports. A better way to enhance this research would be to add to it a backend database which can automatically put these daily logs into monthly attendance summary. This is taken care of by our project since the recognition engine will be interlinked with a detailed reporting module which will help streamline the administrative processes.

2.10 RFID Based Attendance System [10]

Joshi et al. [10] in the last paper of this review dwells upon the hardware-based approach that is based upon the Radio frequency Identification (RFID) technology implementation. The process includes issuing each student with an individual RFID tag, which in most cases is embedded on the ID card, where he or she is required to scan a reader mounted at the entrance of the classroom. Such a system is lauded due to its high speed and low computation cost because it does not demand the high level of processing power that is required in image or biometric analysis. The authors prove that an RFID-based solution is able to scan a complete class of sixty students within less than two minutes, thus it is a highly effective solution in large-scale implementation where cost-related issues are a key factor. The study focuses on the long-life span and convenience that is associated with physical tag-based systems as opposed to sophisticated software-dependent procedures.

Although RFID is efficient, the biggest gap in the research is the impossibility to avoid proxy attendance. The system does not monitor the actual person that is carrying the card it merely monitors the card and as such a single student can easily carry and swipe several cards on behalf of his/her absent friends and this is totally against the integrity of the data. Moreover, the research provides no analytics other than a mere date of a card scan. In order to enhance this model, there is a possibility of using a multi-factor authentication method and this may be a combination of the RFID scan with a second biometric verification (or a localized

mobile verification). The system would also have a cloud based dashboard to see the attendance data in time. Our project will be out of the weaknesses of RFID because we will be doing it in a more safe, biometric centric method, whilst still achieving the high speed data processing objectives as discussed in this study.

Table 2.1 Summary of Literature reviews

Sl No	Author and Year	Concept Used	Method or Approach	Limitation Identified	Improvement Suggested
1	AI based Face Recognition System 2025	Real time face recognition	Deep learning model used to detect and identify students through camera	Performance reduces in low light and when faces are partially covered	Use stronger face detection model and improve lighting adaptability
2	Attendance and Analytics System 2025	Automated data recording with reporting	Attendance stored digitally and reports generated for faculty	Depends on manual input at initial stage	Fully automate attendance capture to avoid wrong data entry
3	Systematic Literature Review 2025	Comparison of existing attendance technologies	Reviewed RFID biometric and mobile based systems	Most systems focus only on recording not analysis	Integrate analytics with attendance data for better decision making
4	Student Monitoring System 2025	Real time monitoring software	Local network based attendance tracking	Limited remote access and no automatic alerts	Use cloud based system with notification support
5	Face Recognition Monitoring 2024	Camera based recognition system	Convolutional neural network for identification	Requires high processing power and storage	Use lightweight processing and better data management
6	Low cost Face Recognition	Haar cascade face detection	Single camera setup with basic recognition	Accuracy reduces with lighting change and head	Replace old algorithms with modern deep

Sl No	Author and Year	Concept Used	Method or Approach	Limitation Identified	Improvement Suggested
	2020		algorithm	movement	learning models
7	Attendance Management System 2021	Face recognition with database integration	Automatic update of digital records	No protection against photo spoofing	Add liveness detection and secure authentication
8	Smart Face Recognition System 2018	Fast processing recognition	OpenCV based high speed identification	Reduced accuracy in crowded classrooms	Add multi frame tracking and improve recognition stability
9	Face Detection and Recognition 2024	Two stage detection process	Face detection followed by recognition	No historical tracking to confirm identity	Add time based filtering to stabilize recognition
10	RFID Attendance System 2021	RFID card scanning	Tag based identification at classroom entry	Proxy attendance possible as only card is scanned	Combine RFID with biometric or identity verification

The literature review in Table 2.1 indicates that various scholars have attempted to enhance attendance systems with the use of various technologies like face recognition and RFID. The majority of the new works are oriented to the automation of the recording process and the minimalization of the manual work. Nonetheless, most of these systems continue to experience such practical challenges as lighting problems, the price of hardware, proxy attendance, and the absence of appropriate analytics. It is also noted that some of the systems primarily focus on capturing attendance but fail to really analyse this data. Based on such observations, our project is to associate secure attendance recording with analytical features such that the system would be more helpful in academic monitoring and decision making.

Chapter 3

Methodology

3.1 Selected Methodology -V-Model (Verification and Validation Model)

To develop the system of Automated Student Attendance Monitoring and Analytics, the V Model methodology was chosen. The reason why this model was selected is because it offers a clear outline of development activities as well as testing activities. As we have to create a project that requires system design, database implementation, user interface development, and analytical processing, we needed to use a structured approach, which would help us to make sure that all the stages would be properly planned and verified.

The V Model is also applicable in the projects where in the start we have the clearly defined requirements. The problem statement was clearly understood in our case. The objectives were to automate the attendance system, minimize the manual error, eliminate proxy attendance, and produce valuable analytics. These were constant demands and could not vary as often hence the V Model was suitable.

The other aspect that led to the choice of this methodology is the fact that each phase of development is related to a phase of testing. This was used to confirm every module before proceeding to the next level. As an illustration, the login module was created and tested first before it was incorporated in the attendance module. This minimized mistakes and increased reliability of the system.

The V Model was well organized and enabled us to take one step after another in the requirement analysis process as we proceeded with the system validation. It also made sure that the end system achieved what was targeted at the start of the project..

3.2 Mapping of V Model to the Proposed System

The requirement analysis was conducted in the first stage. We researched on the available attendance system in colleges and noted its shortcomings. The functional requirements were established based on the secure login, attendance record, analytics dashboard, and role based access. Other non functional requirements like performance, security and ease of use were also discovered.

During the stage of system design, the general architecture was ready. The project was partitioned into various modules which included Admin, Faculty, Student and Analytics. The database schema was made with the purpose of recording/storing the student information, attendance records, and reporting. The inter-module flow of data was to be planned.

The elaboration phase was concerned with the inner logic of every module. To illustrate this, we stipulated how face recognition identifies and authenticates student identity, the way attendance percentages are computed and alerts raised on low attendance.

Python was used to code during implementation with necessary libraries. The face recognition was done using OpenCV. Using web technologies, user interface was built and the data storage and authentication were done with the help of back-end logic.

On the testing part, individual modules like the validation of the login, entry of attendance and generating of reports were unit tested. Integration testing was done to make sure that the modules integrated effectively. Lastly, the system testing helped to ensure that the entire application was as per the original objectives.

This hierarchical mapping provided it with a sense of sanity during the development process and that the end system would work properly.

Table 3.1 Mapping of V Model to Project Phases

Development Phase	Activities in Our Project	Corresponding Testing Phase
Requirement Analysis	Identified attendance issues and defined system requirements	System Validation
System Design	Designed modules and database structure	Integration Testing
Detailed Design	Defined internal logic of face recognition and analytics	Unit Testing
Implementation	Coding using Python and OpenCV	Unit Testing
Deployment	Final integration and system setup	System Testing

Table 3.1 presents the way that every phase of development in the V Model has a testing phase associated with it. This was to make sure that mistakes are detected on time and the system is stable during its development.

3.3 Process Flow of the Proposed System

The suggested attendance system adheres to a systematic order of the operations beginning with the authentication of the user up to the generation of reports. It starts with a user entering into the system using valid credentials. Depending on the assigned role, the system will allow one to access various modules like Admin, Faculty, or Student dashboard.

Once the faculty member has logged in, the attendance session is started. The system turns on the camera and takes pictures of the students in the real time. The images taken are stored and then taken through the face recognition module which compares the images with the student database. In case a match is identified, an automatic marking of attendance is performed and it is stored in the database with date and time.

Once attendance is taken then the system will update the percentage attendance of every student. This data is then processed by the analytics module to create reports like daily attendance summary, subject wise attendance and shortage alerts. Students are able to log in to check their attendance, whereas administrators can view consolidated reports to monitor purposes.

Should any student not achieve the needed attendance percentage, then the system can draw attention of the same in the dashboard. This comes into play to identify lack of attendance early as opposed to end of semester.

In general, the process of transferring between the login to the attendance capture, data storage, calculating the percentage, and generating the report is continuous and systematic.

3.4 V-Model Diagram (Process Representation)

V Model representation gives the visual explanation of the relationship between the development activities and testing activities in this project. The stages of development are organized in a decreasing fashion on the left hand side of the model. These processes are requirement analysis, system design, detailed design and implementation. Every step is dedicated to the construction of the particular component of the system in an orderly manner.

On the right hand side of the model, there are testing stages that are in an ascending order. Unit testing is used to test individual modules like the login system and attendance module. Integration testing is to make sure that modules are properly working together. System testing checks the whole application to the original requirements. The V shape structure indicates that there is a corresponding testing activity of each of the development activities. This will make sure that mistakes are detected early and not at the concluding development stages. With this model, the project was able to stay on course, minimize confusion and have the final system achieving the set objectives.

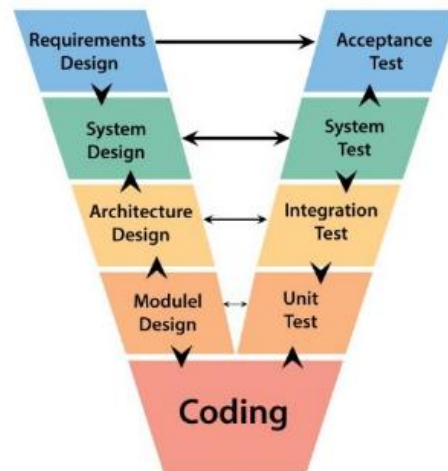


Fig 3.1 The V model methodology

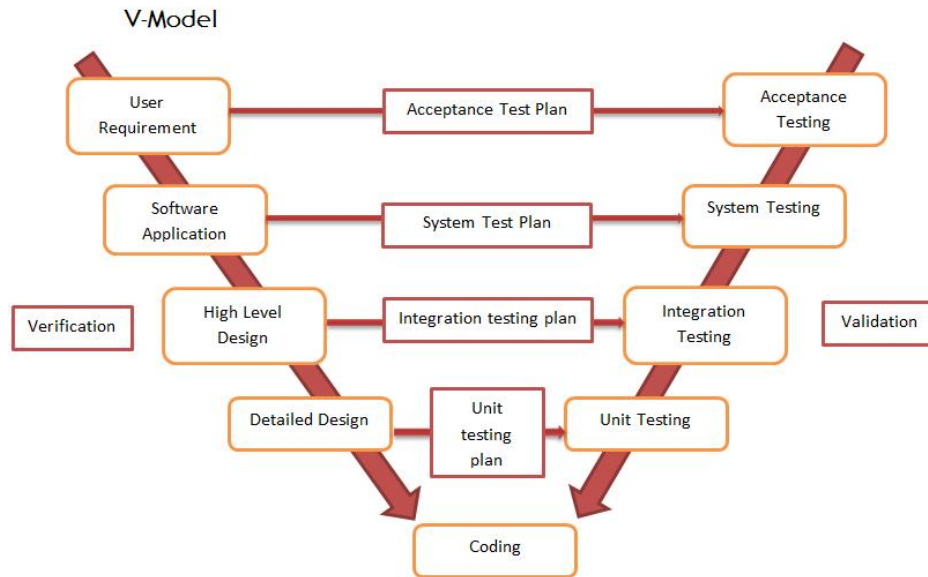


Fig 3.2 Another example of the V model methodology

Fig 3.1 and Fig 3.2 depict V Model representation that will be adhered to in the project. The left side is associated with development phases and the right one is associated with testing phases. This structure will provide adequate checking and validation during development process.

Chapter 4

Project Management

4.1 Project timeline

The project timeline was divided into two main stages. The first stage focused on planning activities such as requirement gathering, literature review, and system design. The second stage focused on development, testing, and final documentation.

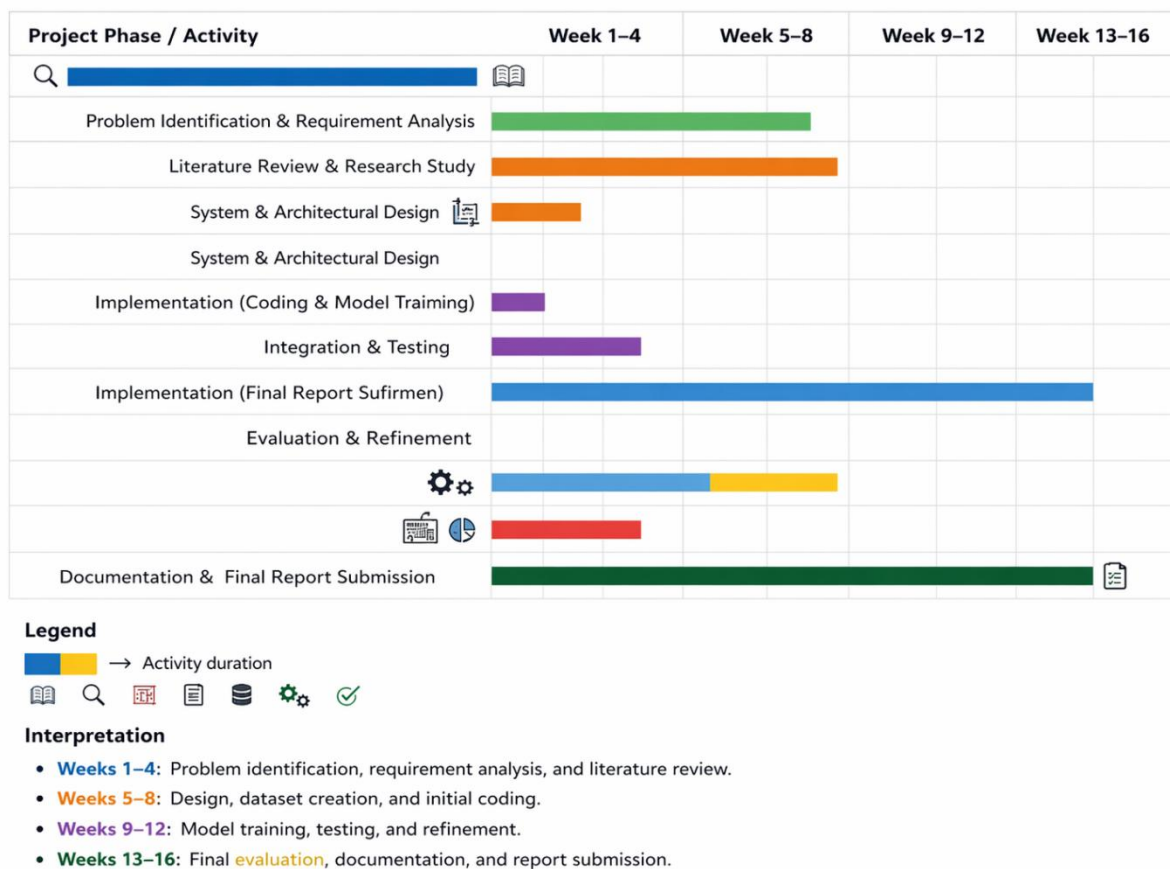


Fig 4.1 Project Timeline (Gantt Chart Representation)

It ensures systematic progress with defined milestones related to the gathering of requirements, design, coding, testing, and final documentation as shown in Fig 4.1

4.2 Risk Analysis

Risk analysis allows the comprehension of what challenges may come up during the implementation process. The section provides an overview through both PESTLE analysis and a Risk Matrix in identifying various technical, managerial, and operational threats as shown in Table 4.1.

Table 4.1 Project implementation timeline

	Major Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Simulation																
Unit																
Integrated																
Hardware implementation																
Software																
Testing *																
Critical Evaluation **																
Social, Ethical, Legal, and Sustainability Report																
Final report																
* Develop test plan, Identifying test points Black box testing (positive, negative, boundary), White box testing (Control flow, Data flow, Branch, Path) Hardware testing - Unit Testing, Integrated testing Software testing System testing - Validation (dynamic, testing user requirements) Tabulating test results		** Identify the Hardware functional units - Sensors, Input devices, Micro controllers, Actuators, Output devices, Interface circuits, Signal conditioning circuits, Driver circuits Identify the Software functional units - Software component, Initializing, Acquiring, Processing, Data Logging, Controlling, Indicating Discuss the properties, issues, constraints of each functional units, Working principle, Signal type (digital or analog), Signal conditioning (signal level, noise, signal conversion), Latency, Linearity, Accuracy Discuss the aspects to improve each functional units, Reliability, Power aware, Interrupt driven, Precise timing (Real time), Indicate output, Meet standards, Safety														

4.2.1 PESTLE Analysis

P Political	E Economic	S Societal	T Technological	E Environmental	L Legal
- Taxation policies - Trade restrictions - Tariffs - Political stability	- Interest rates - Exchange rates - Inflation rates - Raw material costs - Employment or unemployment rates	- Population growth - Age distribution - Education levels - Cultural needs - Changes in lifestyle	- Technology development - Automation - R&D	- Climate - Weather - Resource consumption - Waste emission	- Discrimination law - Consumer law - Antitrust law - Employment law - Health and safety law

Fig 4.2 Pestle Analysis

This Fig 4.2 shows the PESTLE framework applied to the project, illustrating political, economic, social, technological, legal, and environmental factors affecting development.

Table 4.2 PESTLE Analysis

Factor	Description	Impact on Project
Political	Deployment of the system in institutions may be influenced by government regulations related to student data privacy and digital systems.	Medium
Economic	Budget constraints may affect the quality of hardware such as camera systems or server capacity.	Medium
Social	Students and faculty acceptance of automated attendance and face recognition may vary based on awareness and privacy concerns.	High
Technical	The system depends on face recognition accuracy and database stability which require continuous updates and maintenance.	High
Legal Compliance	Compliance with data protection laws such as IT Act 2000 and institutional data policies is mandatory.	High
Environmental	The system reduces paper usage and has minimal environmental impact since it is software based.	Low

The above table 4.2 identifies possible external factors that may influence the project. Social acceptance and legal compliance are considered high impact areas because the system deals with student identity and personal data. Proper data handling and transparency are necessary to reduce these risks.

4.2.2 Risk Matrix

Table 4.3 Risk Matrix

Risk Description	Probability	Impact	Level	Mitigation Strategy
Face recognition may fail due to lighting variations or	High	High	Critical	Use image pre-processing techniques and collect images under different lighting

Risk Description	Probability	Impact	Level	Mitigation Strategy
partial face visibility				conditions
Database corruption or data loss	Medium	High	High	Maintain regular backups and use secure database configuration
Unauthorized system access	Medium	High	High	Implement strong authentication and role based access control
System performance slowdown during peak usage	Medium	Medium	Moderate	Optimize code and monitor server performance
Resistance from users unfamiliar with technology	Low	Medium	Low	Provide awareness and simple user interface

The risk matrix Table 4.3 helps identify technical and operational risks that may affect system performance. Face recognition accuracy is considered critical because it directly affects attendance reliability. Proper testing and secure system design help reduce these risks.

Project Phase Risk Matrix							Probability					
							Almost Impossible (1)	Not likely to occur (2)	Could occur (3)	Known to occur (4)	Common occurrence (5)	
Regimes		Health and Safety	Environmental Impacts	Financial & Asset Loss	Reputational Damage	Production / Projects	Information Technology	Occurs less than once in 10 000 years	Occurs once in 1 000 to 10 000 years	Occurs once in 100 to 1 000 years	Occurs once in 10 to 100 years	Occurs once in 1 to 10 years
Potential Consequences	Catastrophic (5)	One or more fatalities. Irreversible health problems for employees and/or community	On or off-site spill causing groundwater pollution, with detrimental long-term effects	Severe financial loss or asset replacement cost impact. (> US\$ 2 million)	International loss of reputation / Damaging International TV exposure with impact	Indefinite cessation of production activity / Extended project schedule slip of > 75% of plan	Significant failure and operational downtime with permanent loss of critical data integrity.	5	10	15	20	25
	Major (4)	Partial, or medium-term, disabilities or major health problems for employees and/or community	Off-site release, contained & medium-term effects on community health and/or groundwater	Major financial loss or asset cost impact. (> US\$ 1 million < US\$ 2 million)	National loss of reputation / Damaging National TV exposure with impact on customers	Long-term production cutback / Major project schedule slip of 40 to 75% of plan.	System failure and operational downtime, with loss of critical data integrity and/or confidentiality.	4	8	12	16	20
	Moderate (3)	Lost-time injuries or potential medium-term health problems for employees and/or community	On site release, contained & restored, with medium-term effects on employees/ groundwater	Moderate financial loss or asset cost impact. (> US\$ 100 000 < US\$ 1 million)	Regional loss of reputation / Local radio & newspaper reports impacting suppliers/customers	Medium-term production cutback / Project schedule slip of 20 to 40% of plan.	System downtime with operational impact / restricted loss of data integrity / confidentiality	3	6	9	12	15
	Minor (2)	Minor, very short-term health concerns or Recordable Injury cases.	On site release, immediately contained & restored, with short-term effects	Tolerable financial loss or asset cost impact. (> US\$ 10 000 < US\$ 100 000)	Loss of regional reputation by word of mouth re. safety performance & treatment of workers	Short-term production cutback / Minor project schedule slip of 10 to 20% of plan	Limited downtime, recoverable data loss with limited operational impact, no security breach	2	4	6	8	10
	Insignificant (1)	Inherently safe, unlikely to cause health problems. First aid injuries.	Minor localised spill with insignificant effects on employees and/or community	Relatively low financial loss or asset cost impact. (< US\$ 10 000)	Unsubstantiated rumours with light to moderate impact on reputation	Very short-term production cutback / schedule slip of up to 10% of plan	Limited downtime, recoverable data loss, workaround possible, no security breach	1	2	3	4	5
		Low risk		Medium risk		Significant risk		High risk				

Fig 4.3 Risk distribution by category.

The Fig 4.3 represents distribution of identified risks based on their impact levels.

4.3 Project Budget

The project mainly involves software development with minimal hardware requirements. Since open source tools are used, the overall cost remains moderate. The estimated budget includes development resources, camera equipment, and documentation expenses as shown in Table 4.4.

Table 4.4 Estimated Project Budget

Resource or Task	Details	Estimated Cost INR
Development System	Laptop or PC for coding and testing	60000
Software Tools	Python, OpenCV, Database tools open source	0
Camera Module	Web camera for face capture	2500

Resource or Task	Details	Estimated Cost INR
Cloud Storage or Hosting	Data backup and storage	3000
Internet and Utility Costs	Required for development and testing	1500
Documentation and Printing	Final report and printing	2000
Contingency	Miscellaneous expenses	3000
Total Estimated Cost		72000

4.4 Summary

The project management activities were relevant in the accomplishment of this project in an organized manner. The development process was organized and manageable by broken down into planning and implementation. An early approach to detecting the risks was used to mitigate significant technical and operational problems. The budget estimation was done in such a way that resources were efficiently utilized with no unnecessary expenditure. In general, good planning, risk management, and cost management have assisted in meeting the project goals within the allocated time as well as ensured that the systems are of good quality and reliable.

Chapter 5

Analysis and Design

5.1 Requirements

It was also necessary to clearly comprehend what the Automated Student Attendance Monitoring and Analytics System should do before designing it. This phase was devoted to the determination of the purpose, the behavior, constraints, and the expectations of the system users. The requirements were categorized into hardware requirements, software requirements, data requirements and security requirements.

The primary aim of the system is to automate the process of attendance recording with the help of face recognition and analytics reports so as to improve academic monitoring. The system is expected to minimize the manual workload and deliver the correct attendance statistics in real time.

5.1.1 Functional Requirements

Functional requirements provide an account of what the system is supposed to do. The main role of the system is to automatically take records of the attendance of the students through face recognition technology. It should enable faculty members to start an attendance session, take a picture of students, ensure identification, and save attendance information in the database. It must also compute the percentage of attendance and come up with reports like the daily attendance summary and shortage lists. The administrator should be able to access the student records, faculty accounts, and system data. The students need to log in and see their individual attendance status.

5.1.2 Non Functional Requirements

Non functional requirements are those that specify the quality and performance requirements of the system. The attendance system should be able to give correct recognition results during normal classroom conditions. It must be able to process data without much hustle. The interface to be used must be user friendly in order to enable faculty members to use it without technical challenge. The system should also provide data security and keep the information of the students secret. Moreover, the system ought to be scalable such that it can accommodate high populations of students should it be applied in several departments.

5.1.3 Hardware Requirements

The system will need a simple computer or laptop with a high processing power to execute face recognition codes. Web camera will be necessary to take pictures of students in the time of attendance. These conditions must be appropriate in terms of light in order to enhance recognition. The hardware parts must be of good quality and be compatible with the operating environment in which it will be developed.

5.1.4 Software Requirements

The system is written in Python programming language and with other supporting library of face recognition and database management. OpenCV processes are utilized in image processing and face detection. A database management system will be necessary to maintain the student information and their attendance records. Web technologies are used to design the front end interface so that it offers user friendly dashboards to the faculty, students, and administrators.

5.1.5 Data and Security Requirements

The system needs to gather student details in the process of enrolling them such as name, roll number, and a sample of their face. They should be in the form of attendance, date, subject, and time. There should be security measures to ensure that data is not accessed by the unauthorized personnel. Role based authentication should be able to make sure that only the relevant individuals make amendments or see sensitive details. Backups must be conducted regularly in order to avoid loss of data.

Table 5.1 Summary of System Requirements

Category	Description
Purpose	To automate attendance recording and generate analytical reports
Behavior	System captures student face, verifies identity, and records attendance automatically
System Management	Admin should manage student data, faculty accounts, and reports
Data Analysis	System should calculate attendance percentage and generate

Category	Description
	summaries
Application Deployment	Web based system accessible within institution network
Security	Secure login with role based access control
Storage	Attendance data stored securely in database
Notification	Alert when attendance falls below required percentage

Table 5.1 summarizes the key requirements of the proposed attendance system in a structured manner. The main purpose of the system is to automate the process of recording attendance and to provide useful analytical reports that help in monitoring student performance. The system is designed to capture student faces using a camera, verify their identity, and record attendance automatically without manual intervention.

5.2 Block Diagram

The functional block diagram will be used to portray the general operation of the Automated Student Attendance Monitoring and Analytics System. It illustrates the way input information is manipulated and transformed into a useful output by various functional units. The system will be broken down into input, processing, storage, and output blocks.

The system begins with user login whereby the members of the faculty authenticate themselves before making an attendance. After the session has started, the camera records the images of the students in real time. The face detection unit will discover the faces in the frame and forward them to the recognition unit where a verification will be done. In case the identity is matched with information in the registered database, the attendance processing unit tags the student as present and records the data in the database. The analytics engine then uses the stored data to determine the percentages of attendance and shortages. Lastly, user dashboards and reports are used to provide the outcomes and view attendance records easily.

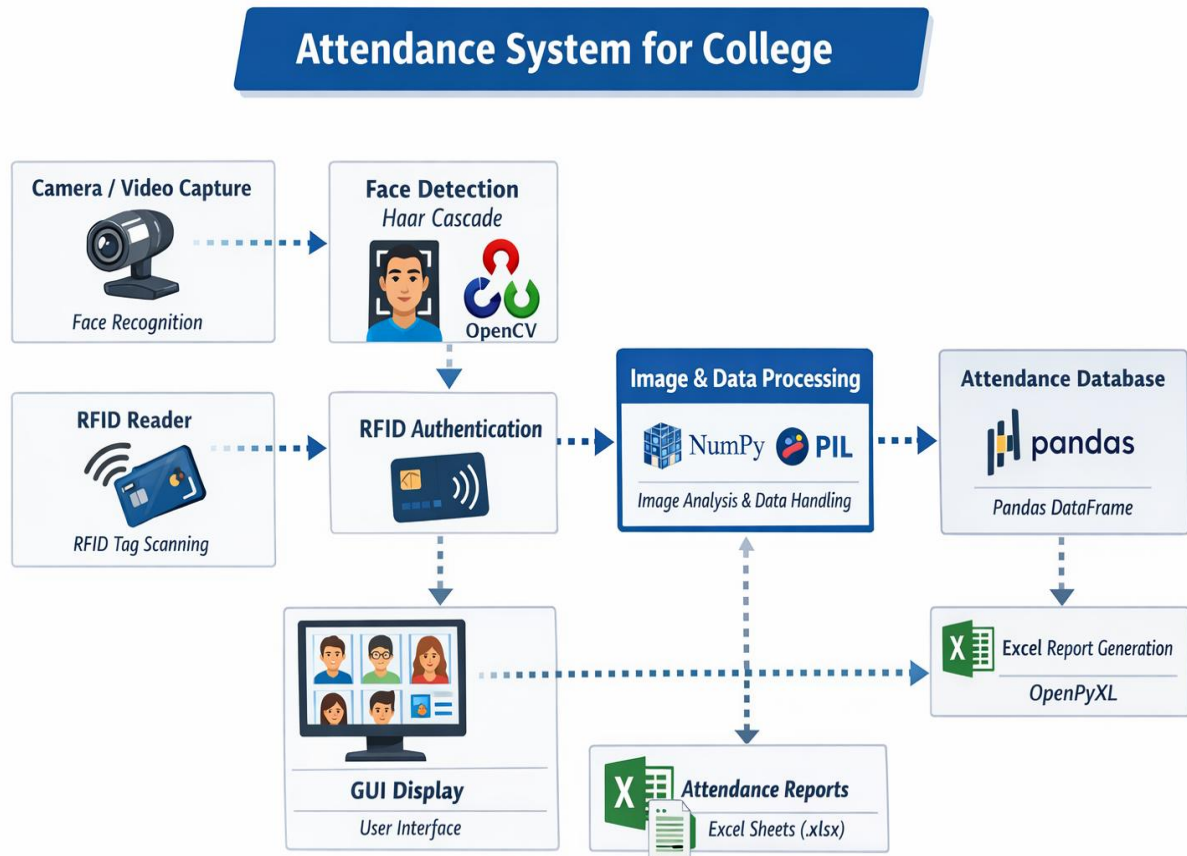


Fig 5.1 Functional block Diagram

Fig 5.1 represents proposed system functional block diagram. The input begins with the user authentication and capturing the image with the help of the camera. The collected pictures are worked out using the face recognition module that verifies the identity of the students against the records in the store. The attendance is then checked and recorded into the database. The data obtained from analytics engine digs are analyzed and used to calculate the percentage of attendance and generate reports. The overall product is also displayed through faculty, students and administration dashboards.

5.3 System Flow Chart

The system flow chart describes how the Automated Student Attendance Monitoring and Analytics System does its operations. It gives a clear picture of the way the system operates between the state of initiation and recording attendance and generating reports.

The operation starts with the opening up of the system. The system is accessed by the user who is normally a faculty member with valid credentials. When the one logs in, the attendance session is triggered. The camera is turned on to take the picture of students at the time of the class. Once the photos are taken, the system does face detection to recognize the human face in the picture. All the faces that are detected are verified against registered student database. In the event of a match, the attendance is recorded as present and saved in the database together with date and time. In case of no match, then the system disregards the entry or sets up an alert to review it later, after all the faces are seen, the system automatically updates the percentages of those who are present. The analytics module is then used to process the data stored and provide attendance summaries. These summaries are posted in the dashboard of faculty and administrators. Another flow the students can also log in individually to view their attendance records. Once the attendance information has been saved and the reports generated successfully, the flow is complete.

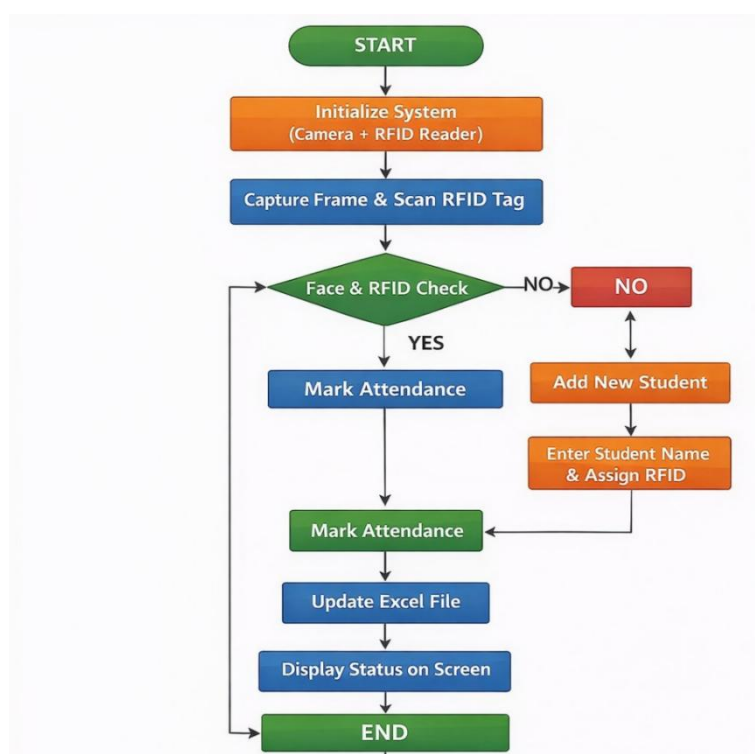


Fig 5.2 System Flow Chart of the Proposed Attendance System

Fig 5.2 is a depiction of the general working flow of the proposed system. It demonstrates the flow of user entry into the system up to capturing attendance, verification and storing the data, calculation of the percentages, and production of a report.

5.4 Choosing Devices

The proposed attendance system incorporates RFID authentication software and hardware based face recognition. This is why it was significant to choose proper equipment to provide the efficient combination of hardware and software elements. This system will need a RFID processing unit, a camera, and a computer system to execute face recognition algorithms.

5.4.1 Choosing Microcontroller

The micro-controller will play the role of reading and sending RFID card data to the main system. Arduino Uno was chosen because the project will only need a simple serial communication and a steady control over the hardware.

Table 5.2 Comparison of Microcontrollers

Feature	Arduino Uno	ESP32	Raspberry Pi
Clock Speed	16 MHz	240 MHz	1.5 GHz
RAM	2 KB	520 KB	2 GB
Operating System	No	No	Yes
Suitable for RFID	Yes	Yes	Yes
Suitable for Face Recognition	No	No	Yes
Cost	Low	Moderate	High

Table 5.2 presents a comparison of different microcontrollers considered for the system, namely Arduino Uno, ESP32, and Raspberry Pi. The comparison highlights important factors such as clock speed, memory, operating system support, and suitability for different tasks. Arduino Uno has a lower clock speed and very limited memory, but it is simple, cost effective, and suitable for handling RFID operations in this project.

5.4.2 Choosing Camera

The camera is used to capture student images for face recognition. Image clarity and stability are important for accurate identification.

Table 5.3 Camera Comparison

Feature	USB Web Camera	Laptop Built in Camera
Resolution	720p to 1080p	720p
Ease of Setup	Plug and Use	Already Available
Image Quality	Good	Moderate
Flexibility	Can be positioned freely	Fixed position

Table 5.3 compares the USB web camera and the laptop built in camera based on features such as resolution, ease of setup, image quality, and flexibility. The USB web camera provides better resolution and image quality, which improves face recognition accuracy. It is also more flexible as it can be positioned according to the classroom setup.

5.4.3 Choosing RFID Module

RFID is used as an additional authentication mechanism to reduce proxy attendance. The module must be compatible with Arduino and support short range reading.

Table 5.4 RFID Module Comparison

Feature	RC522	PN532
Operating Frequency	13.56 MHz	13.56 MHz
Communication	SPI	SPI I2C UART
Reading Range	3 to 5 cm	5 to 7 cm
Cost	Low	Moderate
Compatibility with Arduino	Yes	Yes

Table 5.4 compares the RC522 and PN532 RFID modules based on key features such as communication methods, reading range, cost, and compatibility. Both modules operate at the same frequency and are compatible with Arduino, but the PN532 offers more communication options and a slightly longer reading range. However, the RC522 is more cost effective and

sufficient for short range attendance applications. Based on this comparison, RC522 is preferred for the proposed system due to its simplicity and low cost.

5.5 Designing Units

This proposed attendance system is broken down into various functional units in order to have a given task of each component. Subdivision of the system into smaller units facilitates easier development and it also aids in testing the parts individually. The key components in the system are face recognition unit, RFID authentication unit, database management unit, and analytics unit.

5.5.1 Face Recognition Unit

The face recognition unit will be in charge of students identification in the attendance session. Upon the system initiation, the camera takes photos of the students on the classroom. The processing of these images will be done by applying face detection algorithm that detects the human faces in the image captured. After identification of the face, the system will match the face with the student face database that is stored to confirm identity.

In case the face that is captured is similar to a student record in a database, the system validates the identity and the student is registered as present. The process is automatic and does not require the faculty to input into it manually. This unit is determined by the quality of images, light conditions and the quantity of images in the training set which determines the accuracy of this unit.

5.5.2 RFID Authentication Unit

The RFID authentication component offers another authentication to curtail chances of proxy attendance. Every student is issued with an RFID card that has a unique identity number. When the card is swiped around the RFID reader, the reader reads the card ID and transmits the information to the Arduino micro-controller.

The Arduino processes the card ID obtained and transmits it to the computer system via the serial communication. The system will then authenticate the card ID with those of registered students. In case the card is valid then the authentication is made and the attendance record is updated. This unit makes sure that the registration of attendance is only done by authorized students.

5.5.3 Database Management Unit

It is the mandate of database management unit to store and organize all data in the system. It holds the registration information of students, samples of the face image, RFID card IDs, and information on the attendance of students on a daily basis. The database enables the system to access information fast when conducting face recognition and RFID verification.

When the attendance is noted, it is always recorded with the date and time. There is also the ability to generate reports and analyze attendance in the database. Correct database design guarantees that the information is always uniform and convenient to handle.

5.5.4 Analytics and Reporting Unit

The analytics unit runs attendance records stored in the database and produces valuable information. It calculates the percentage attendance of individual students and determines those students whose attendance is at a level of lower than the set required percentage. Various kinds of reports could be produced by the system including daily attendance summary, attendance by subject, and shortage list.

These reports are represented by dashboards that are available to faculty and administrators. The students may log in to know their attendance status as well. The analytics team assists institutions to track the student attendance and implement required measures in instances where attendance is low.

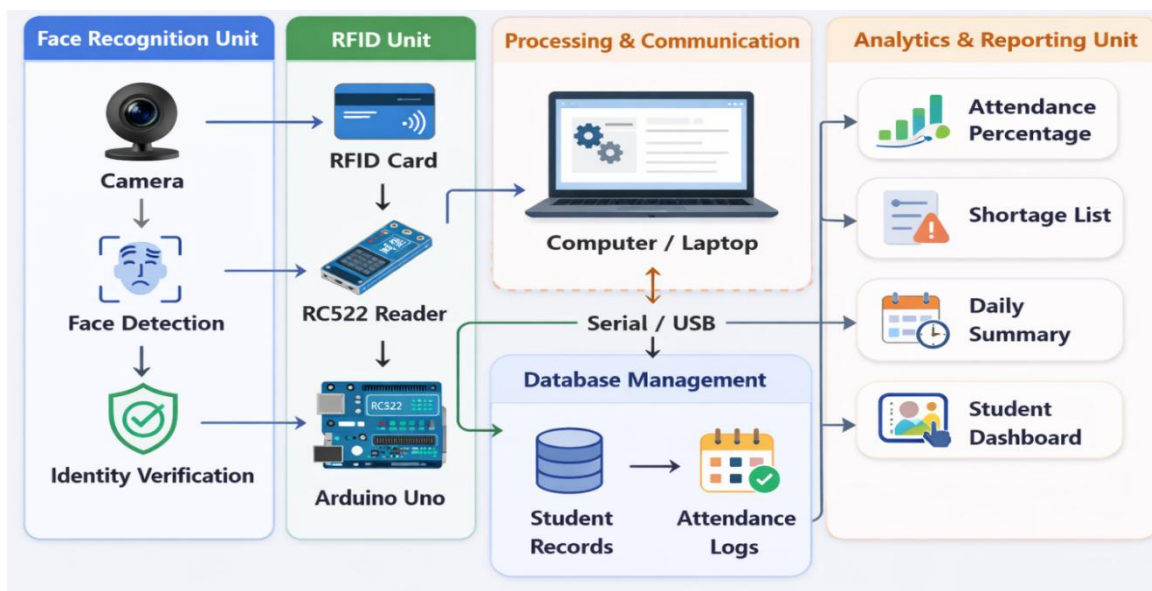


Fig 5.3 Functional Unit Interaction in the Attendance System

Fig 5.3 demonstrates the interaction of various units of the system to one another. Student identity is obtained by face recognition unit and RFID authentication unit. The information is then worked upon and stored in the database management unit. This data is retrieved in the analytics unit and uses it to produce reports and attendance summaries.

5.6 Standards

Standards are significant in the way various parts of a system operate in the right manner. In the proposed attendance system, there are various technical standards that are taken into consideration in achieving a form of reliable communication, exchange of data in addition to the security of the system. They are standards that are utilized to maintain compatibility in both hardware and software modules as well as database systems.

The system features normal serial communication between the Arduino microcontroller and the computer system. The RFID data that the reader reads can be relayed to the processing unit to be verified by this kind of communication. The camera and Arduino device are connected to the computer by the use of standard USB communication.

In case of data management, the student records and attendance information are stored in the database in form of structured data format. This guarantees ease with which the data can be accessed and manipulated by the analytics module. The storage of data is done in accordance to basic database management to ensure accuracy and consistency.

Another standard of importance that is taken into consideration in the system is security. The authentication of the users is done to make sure that the system is accessed by only authorized users. Role based access control enables the administrators, faculty and students to have access to features they need. This assists in securing confidential student data.

Adherence to these standards assures that the system is stable, secure and is compatible with other hardware and software.

Table 5.5 Standards Used in the System

Standard Type	Description	Application in Project
Communication Standard	Serial communication through USB	Data transfer between Arduino and computer

Standard Type	Description	Application in Project
Data Format Standard	Structured database records	Storage of student and attendance data
Security Standard	User authentication and access control	Prevent unauthorized access
Hardware Interface	USB interface for devices	Connection of camera and RFID reader
Database Standard	Relational database structure	Efficient storage and retrieval of attendance data

The key standards that were taken into account in the system are summarized in Table 5.5. The standards provide easy communication amid hardware components and secure management of student information.

5.7 Mapping with IoTWF Reference Model Layers

The reference model of the IoT World Forum offers a layered framework that assists in the perception of the way various elements of an IoT system interact with one another. Even though the suggested attendance system is primarily a software based application, cameras and RFID modules can be used to interpret the system as a part of the IoT reference framework. Mapping the system to those layers assists in the analysis of how data passes between physical devices and an analytical application.

The bottom layer in the proposed system is the physical devices which are camera and RFID reader. These devices record the identity of the students during the attendance sessions. The connectivity layer consists of dealing with communication between the computer system and the hardware devices using the USB and serial communication. Processing layer is involved with face recognition and RFID verification.

The information which has been captured is stored in the database using the data accumulation layer. The information is then processed by the data abstraction layer once it has been stored and then readied in order to be analysed. The application layer is used to create attendance reports and analytics dashboards. Lastly, the collaboration layer signifies

the communication between the administrators, faculty personnel, and students who utilize the system to track attendance.

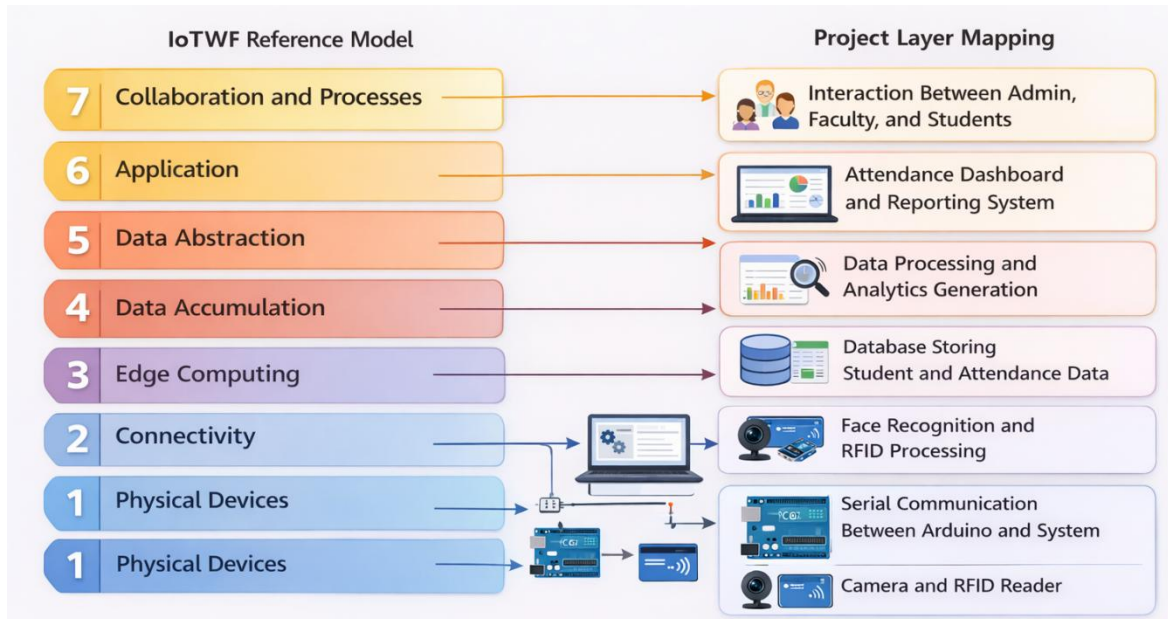


Fig 5.4 IoT World Forum Reference Model Layers

Fig 5.4 shows the reference model of IoT World Forum which has various layers beginning with the physical devices through to application and collaboration layer. The layers have certain functions to carry out in terms of data and system operations.

Table 5.6 Mapping Project Layers with IoTWF Reference Model

Layer	IoTWF Reference Model	Project Layer Mapping	Security Consideration
7	Collaboration and Processes	Interaction between admin, faculty, and students	Role based access control
6	Application	Attendance dashboard and reporting system	User authentication
5	Data Abstraction	Data processing and analytics generation	Secure data processing
4	Data Accumulation	Database storing student and attendance data	Database security

Layer	IoTWF Reference Model	Project Layer Mapping	Security Consideration
3	Edge Computing	Face recognition and RFID processing	Device level verification
2	Connectivity	Serial communication between Arduino and system	Secure device connection
1	Physical Devices	Camera and RFID reader	Controlled hardware access

Table 5.6 presents the mapping of the components of the attendance system with the layers of the IoT World Forum reference model. The mapping assists in describing the flow of data between the physical devices and application level services. It also brings out security controls that are implemented at various points of the system.

5.8 Domain Model Specification

The domain model is the description of key entities and relations of the Automated Student Attendance Monitoring and Analytics System. It gives one a conceptual picture of the interaction between various components in the system. The model recognizes the physical objects that are found in the environment and the digital equivalents of the objects in the system.

The main physical entities in the proposed attendance system are students. In the system database, there is a virtual entity of each student. The virtual entity contains the information including the name of a student, registration number, face image dataset and RFID identification number. The camera and RFID reader are some of the devices that interact with these entities to record information about identity.

The system resources consist of software modules that analyze the data that is captured and save it on the database. Services have a responsibility of availing this information to dashboards and reports. These services enable the administrators, the faculty members, and the students to communicate with the system and track the attendance records.

Table 5.7 Description of Domain Model

Component	Description
Physical Entity	Students present in the classroom whose attendance needs to be recorded
Virtual Entity	Digital representation of students including name, ID number, face data, and RFID card details
Device	Camera and RFID reader used to capture identity information
Resource	Software modules such as face recognition algorithm and database system
Service	Attendance marking, report generation, and analytics dashboard

Those involved in the domain model are explained in Table 5.7. These are the aspects that can be used to determine the relationship between real world objects and the digital system that should be used to monitor attendance.

5.9 Communication Model

The system design of the Automated Student Attendance Monitoring and Analytics System involves communication between its various elements. The system operates on request response model of communications wherein a request by a user or device is handled by the system and the corresponding answer is sent back.

The communication in the suggested system will start once the faculty member enters the system and initiates the attendance session. The system sends a request to open the camera and the RFID reader. When the RFID is scanning the card, the camera captures the student image. These are the inputs that are passed to the processing module to be verified.

After the system has collected the information captured, it compares the information to the database in order to confirm the student identity. When the data is a match with the records stored, the system will then respond by marking the attendance and sending the data in the database. The revised attendance is thereafter shown on the dashboard.

This model of request response is used to provide a smooth and effective interaction between the hardware devices, software modules and database. It enables the system to serve each request on its own and give out the correct answers without having any implications on the functioning of the system

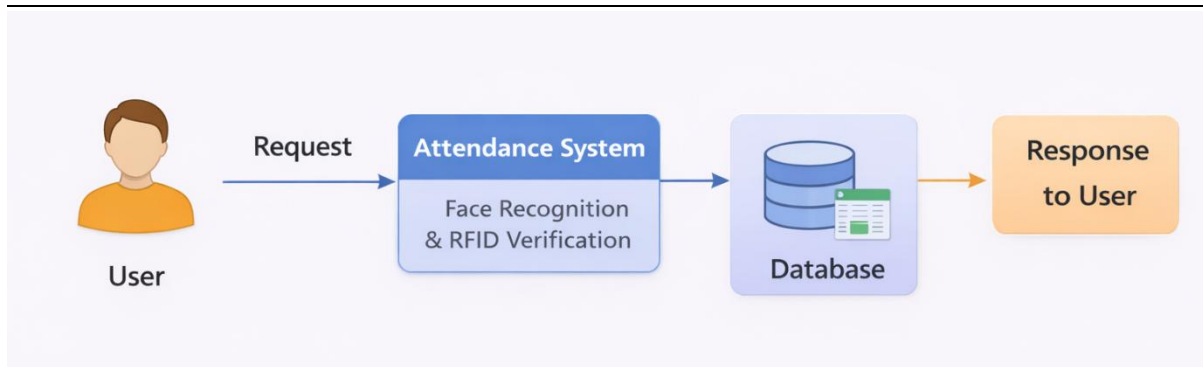


Fig 5.5 Communication Model of the Attendance System

Fig 5.5 indicates the process of communication that is employed within the system. Within an attendance session started, a request is created. The request is processed through the system in a face recognition and RFID verification. The response is saved in the database and the system responds back to the user interface.

Table 5.8 Communication Model Summary

Component	Role in Communication
User Interface	Sends request to start attendance session
Camera and RFID Reader	Provide input data to the system
Processing Module	Verifies identity using face recognition and RFID data
Database	Stores and retrieves student and attendance records
Dashboard	Displays attendance results to users

Table 5.8 represents what is included in the system communication and how it will help break down the requests and responses.

5.10 IoT Deployment Level

The level of IoT deployment defines the interrelation among the devices, processing units, and applications within an IoT based system. Various levels of deployment characterize the way in which data is gathered, processed and stored. The suggested attendance system is based on the deployment scheme according to which student identities are recorded by corporeal devices and processed by the system which will in turn be stored in the database.

The camera and the RFID reader are the devices used in the proposed system as the sensing layer. These gadgets record the faces of students and RFID cards when they are in attendance. The data obtained is forwarded to the processing unit where face recognition and identity identification is done.

The database contains the processed data which has been verified. This information is accessed by the application layer and used to create attendance report and analytics dashboards. The system interface allows faculty members and administrators to see the attendance statistics and students to see their own attendance records.

This deployment setup will guarantee the fact that data processing is performed locally coupled with the database and application services controlling the attendance records and reporting. The architecture offers effective operation of the system as well as preserving data accuracy and security.

Table 5.9 IoT Deployment Components

Component	Role in System
Physical Devices	Camera and RFID reader capture student identity data
Processing System	Performs face recognition and identity verification
Database	Stores student records and attendance data
Application Layer	Displays reports and dashboards for users

5.11 Mapping IoT Deployment Level with Functional View

The mapping of the level of deployment of the IoT and the functional view assists in justifying how the system functions work in the general architecture. The deployment level defines the position of the system component whereas the functional view defines the responsibilities of the system components.

The hardware in the proposed attendance system (including the camera and the RFID reader) are part of the device layer. These machines gather identity data of the students. Connectivity layer, the Arduino micro-controller and the processing system communicate with each other via serial communication.

The processing system is able to carry out functions like face recognition and RFID verification. This is in reference to the service and application functional groups. The student records and attendance data is stored in the database layer and this serves the management functions of the system.

Lastly, the layer of collaboration is the application dashboards which are utilized by administrators including faculty members and students. These dashboard-like systems enable users to engage with the system and see attendance reports as well as see attendance statistics.

The system architecture can be visualized by mapping the functional groups to the deployment structure making the system architecture easier to understand and manage. The layers have certain functions to perform and a role to play in the system as a whole.

Table 5.10 Mapping of Deployment Level and Functional Groups

Deployment Component	Functional Group	Role in System
Camera and RFID Reader	Device	Capture student identity data
Arduino Communication	Communication	Transfer RFID data to system
Processing System	Services	Perform face recognition and identity verification
Database System	Management	Store and manage attendance records

Presidency School of Computer Science and Engineering, Presidency University

Deployment Component	Functional Group	Role in System
Dashboard Interface	Application	Display reports and analytics

Table 5.10 demonstrates the correspondence of functional groups of the system with the deployment components. Such mapping assists in depicting the connection of system structure and system functionality.

5.12 Operational View

The operational perspective explains the way the various parts of the Automated Student Attendance Monitoring and Analytics System would work in the real system. It describes the interaction between devices, processing units, storage systems, and applications to do attendance monitoring.

The RFID reader and the camera are input devices in the proposed system and they are used to record the identity of students. The RFID reader reads the student identification card and the camera takes face pictures. These gadgets are attached to the processing system either by USB or serial interface.

The system carries out its key functions with the help of the processing system. It runs the face recognition algorithm, checks the information of RFID card, and identifies the validity of the student identity. The system logs the attendance information once it has been verified.

Student records and attendance logs are stored in the database server. It has organized data like student data, facial images registered, RFID databases, and records of attendance daily. This data storage is subsequently utilized in the production of attendance report and analytics.

The application layer will have dashboards through which administrators, faculty members and students will interact with the system. Faculty members would be able to start attendance sessions and see reports. Administrators would be able to track attendance data and process system data. Students are able to log in and verify their respective attendance status.

This system of control is effective as it will guarantee that the system is efficient and will have secure storage and convenient access to the attendance data.

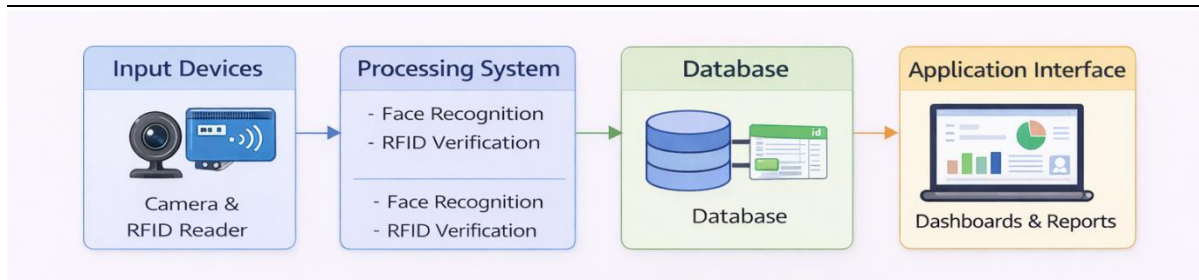


Fig 5.6 Operational View of the Attendance System

Fig 5.6 depicts the working system of the attendance system. The identity information of the student is captured using input devices, the data is verified using the processing system and the database is used to store the attendance record and the application interface is used to have access to reports and dashboards.

5.13 Other Design Aspects

Other than the primary architectural elements, there are other design points that were taken into account in the creation of the Automated Student Attendance Monitoring and Analytics System. All these elements are useful in ensuring the smooth running of the system and the efficient nature of attendance check.

The process specification defines the order of operations on the system. The camera and RFID reader are triggered when a faculty member triggers an attendance session to record student identity information. Face recognition and RFID verification is used to process the captured data. Once the system has been verified, it captures the attendance and enters the data in the database. Data stored is in turn used to create reports and analytics.

Information model is concerned with the way data is structured in the system. Student information, face image collections, RFID identification, and check-in data have been stored in the organized database tables. This is a structured storage that enables the system to retrieve and analyze the information on attendance effectively.

The service specification defines the services that the system offers to various users. Faculty members are able to start attendance sessions and see class attendance reports. Administrators are able to control student records and overall attendance rates. Students have access to the

system to identify their attendance status. These services maintain that the system will accommodate all the necessary academic monitoring activities.

Taking into account these other design characteristics is a certainty that the attendance system will be reliable, organized and able to manage various operational needs.

Table 5.11 Additional Design Components

Design Aspect	Description
Process Specification	Defines steps for attendance capture and verification
Information Model	Describes storage structure for student and attendance data
Service Specification	Defines services provided to faculty, admin, and students
Data Management	Ensures proper storage and retrieval of attendance records
User Interaction	Allows users to access dashboards and reports

Table 5.11 is a summary of the other design considerations incorporated in the system. These attributes will guarantee that the design of the system facilitates high-quality data processing, confidential information storage, and efficient communication with the user.

Chapter 6

Hardware, Software and Simulation

6.1 Hardware

The hardware components are of relevance in the deployment of the Automated Student Attendance Monitoring and Analytics System. This system incorporates physical equipment which captures student identity data and sends it to the processing system. A camera, RFID reader, Arduino micro-controller, and a computer system are the main hardware components of this project.

The camera can be used to take pictures of the faces of students during the attendance session. Face recognition module processes these images to recognize the student. To be able to recognize the camera, image clarity and stable position are required.

Student identification cards are scanned with the RFID reader. Every student is given a special RFID card that carries a special ID. Upon the card being put close to the reader the RFID module reads the card ID and transmits the data to the Arduino micro-controller.

The Arduino micro-controller is an interface between the RFID reader and computer system. It takes the RFID reader card identification data and transmits it to the processing system via a serial communication.

The computer system carries out the core processing functions of face recognition, identity verification, interacting with the database and generating a report. It executes the attending software that has been created.

These hardware parts combined enable the system to record the student identity data and automatically take an attendance.

Table 6.1 Hardware Components Used

Hardware Component	Function
Camera	Captures student facial images
RFID Reader	Reads student RFID identification cards
Arduino Uno	Transfers RFID data to the system
Computer System	Runs face recognition and attendance software

USB Interface	Connects hardware devices to the computer
---------------	---

Table 6.1 shows the hardware components employed in the project and their functions. These elements collaborate to ensure the student identity information is captured and aid in the process of attendance monitoring.

6.2 Software Development Tools

The Automated Student Attendance Monitoring and Analytics System needs to be written, tested, and executed using software tools. The tools assist in the deployment of face recognition, RFID communication and database management, and the development of the application interface. This project has predominantly used open source tools to ensure that the system is flexible and cost effective.

The programming language employed in the system is Python. The reason why Python was chosen is that it has a good image processing and machine learning library support. Integration with hardware devices and databases is also easy.

Face detection and recognition are done with the help of the openCV library. It enables the system to take the images of the camera, identify faces in the captured image and match them with other stored student images to verify the identity.

Arduino IDE is a programming environment that is used to write and upload programs to the Arduino micro-controller. The software is capable of communication between the RFID reader and the computer system. Serial communication is used to transfer the RFID card information that is read by the module into the system.

Student information and attendance records are stored using a database management system. The database stores the student data, facial images, RFID identification, and attendance records.

Visual studio code is a development environment that is used to write and maintain the python source code. During development, this environment offers debugging, code editing and also the application is run in this environment.

Table 6.2 Software Development Tools

Software Tool	Purpose
Python	Main programming language used to develop the system
OpenCV	Image processing and face recognition
Arduino IDE	Programming and communication with Arduino micro-controller
Database System	Storage of student and attendance records
Visual Studio Code	Code development and debugging environment

Table 6.2 contains the key software tools that were utilized in the design of the attendance system. The individual tools can be used in a particular aspect of the system like programming, image processing, hardware communication or database management.

6.3 Software Code

Automated Student Attendance Monitoring and Analytics System is a Python and Arduino programmed software code that was created. Primary face recognition processing and attendance management are done with the help of Python code, whilst RFID card information is read with the help of Arduino code and sent to the computer system.

The software is separated into various modules including face recognition, identity recognition, RFID reading, communication with the database, and record keeping. The modules carry out particular functions and in combination they make sure that the system is able to record attendance automatically and in a more accurate manner.

Code

```
import cv2
```

```
import datetime
```

```
camera = cv2.VideoCapture(0)
```

```
ret, frame = camera.read()
```

```
face_detector = cv2.CascadeClassifier('haarcascade_frontalface_default.xml')
```

```
faces = face_detector.detectMultiScale(frame, 1.3, 5)
```

```
for (x, y, w, h) in faces:
```

```
    student_id = "Detected_Student"
```

```
    time_now = datetime.datetime.now()
```

```
    print("Attendance recorded for", student_id, "at", time_now)
```

```
camera.release()
```

The logic of the attendance system is shown in the above code. A frame of an image is captured by the camera and the faces in the image are identified by the OpenCV library. Upon detection of a face, the system identifies the student and logs in the attendance and the date and time. This information is stored in the database and utilized to produce attendance reports in the entire system.

This is a simplified version of a code snippet that contains the working logic of the system, and the entire implementation involves other modules to verify RFID, store information to the database and also to generate reports.

6.4 Simulation

The Automated Student attendance monitoring and analytics system is tested through simulation to ensure the system works before its full deployment. It makes it possible to test the system under a controlled environment without having to set up large scale hardware. The simulation is used to ensure proper functioning of the face recognition module, RFID authentication module as well as the attendance recording process.

The use of the camera to obtain images of the students and the face recognition algorithm to identify the faces occurred during the simulation process. The images captured were compared with the registered images in the database with the system. In case the face spotted was that of a registered student account, the system was considered present.

Simulation also tested the RFID module. When an RFID card was held close to the reader, the reader was able to receive the card identification number and transmit the information to

the system via the Arduino micro-controller. The RFID identification number was checked against the student database to verify that the student was who he was.

Upon identifying, the system registered attendance data and date and time. The analysis of the stored data was followed by the creation of the attendance reports and analytics. The outcome of the simulation proved that the system could identify student identity and capture attendance of students correctly.

Table 6.3 Simulation Test Results

Test Scenario	Expected Result	Observed Result
Face detection using camera	Student face detected successfully	Face detected correctly
Face recognition verification	Student identity verified	Identity verified
RFID card scanning	Card ID captured correctly	Card ID detected
Attendance recording	Attendance stored in database	Record stored successfully

Table 6.3 displays the result of the simulation testing. All the system components were tested individually to ensure that they were functioning. The findings show that face recognition module, RFID authentication module and the process of counting attendance were functioning well.

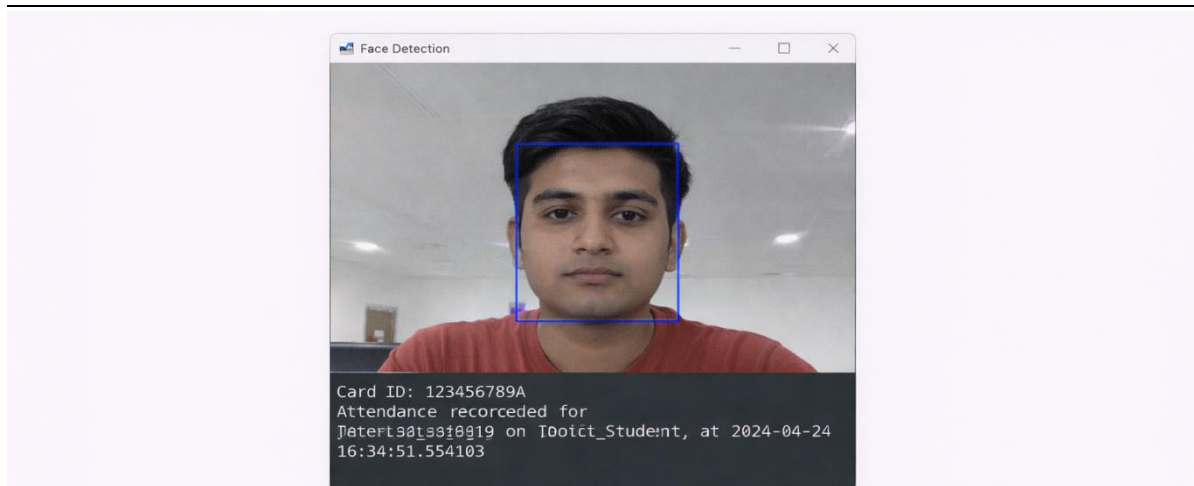


Fig 6.1 Simulation Output of Attendance System

The Fig 6.1 demonstrates the output of the simulation of the attendance system. The system takes the images of the students with the camera, identifies the faces, authenticates and records the attendance in the system.

Chapter 7

Evaluation and Results

The assessment step will be relevant in order to confirm that the Automated Student Attendance Monitoring and Analytics System works and achieves the project goals. At this phase, the various parts of the system including face recognition, RFID authentication, attendance record, and database storage were tried. Findings of these tests are used to identify the reliability and the effectiveness of the system.

7.1 Test points

Test points are some of the features or components of the system that must be tested. All the key functions of the attendance system were acted in a manner that one was tested at a time and this was done to ensure that the system works properly and reliably. The camera was also tested to ensure that it takes pictures properly and that it identifies student faces. The RFID module was provisioned to ensure it reads the student identification card correctly. The face recognition part was tested to determine the ability of the system to make proper matching of the face detected and the registered database. The database section was put to test to make sure that the attendance in the database is inputted and recalled properly. The testing of these separate elements allowed verification that the system is doing all that was needed and then testing the complete system.

Table 7.1 Test Points

Test Component	Description
Camera Module	Capture student images for face detection
Face Recognition	Identify student using stored facial dataset
RFID Authentication	Verify student identity using RFID card
Attendance Recording	Store attendance data with date and time
Database Storage	Save and retrieve student attendance records

The table 7.1 indicates the key components that were tested throughout the system evaluation. All components are major functionality that are needed to automate attendance monitoring.

7.2 Test Plan

The test plan outlines the process that will be used to assess the behavior of the attendance system. This was done in phases whereby every module had been tested before being integrated with those.

First, the camera unit was in test to ascertain that the system was able to record the images. The face detection algorithm was then carried out after the image was successfully captured to ensure that the system was able to detect human faces in the captured image.

Then, the RFID reader was confirmed having scanned student identification cards. The student database was checked to determine whether the system was able to identify registered students according to the card ID.

Lastly, the attendance recording feature was checked through the check of whether the system accurately records the attendance data following the successful identity check.

This method of testing in a structured way was able to help find possible problems and also made sure that every single component was able to perform the functions it was meant to.

7.3 Test Results

The system was tested on the accuracy of face recognition, RFID authentication and attendance recording after the implementation of the test plan. The findings revealed that the system could detect the face of students and authenticate identities through face recognition and RFID authentication.

The system was able to record all the attendance of verified students and save the information in the database and date and time information. The attendance reports that were generated were also true to the data recorded.

The testing exercise helped to confirm that the system could take automated attendance monitoring without significant mistakes.

Table 7.2 Test Results

Test Scenario	Expected Result	Observed Result
Face detection	Detect student face in camera frame	Face detected successfully
Face recognition	Match detected face with student dataset	Student identified correctly
RFID scanning	Read student card identification	Card ID captured successfully
Attendance recording	Store attendance in database	Attendance stored correctly

Table 7.2 presents the comparison of the expected behavior of the system and the results observed during the testing. The findings affirm that the system has worked within the specifications of the design.

7.4 Insights

The results of the evaluation show that the attendance process can be automated by the proposed attendance system. The RFID authentication of the system combined with face recognition enhances system reliability and prevents the possibility of proxy attendance.

The testing process also pointed out the value of the right light conditions in detection of faces. Having the straight camera position aids in increased recognition. RFID authentication also involves an extra verification step that further enhances system reliability.

On the whole, the system was successful in the course of testing and met the primary goal of the student attendance monitoring automation.

Chapter 8

Social, Legal, Ethical, Sustainability and Safety aspects

The promotion and implementation of the technological systems must be based on the technical performance, as well as on the social, legal, ethical, sustainability and safety factors. These points should be taken into consideration because the Automated Student Attendance Monitoring and Analytics System gathers and processes data of student identity. These factors can be taken into consideration to make sure that the system is responsible, secure, and useful to the users and institutions.

8.1 Social Aspects

The suggested attendance system is capable of influencing the academic environment positively. Automating attendance tracking helps to save time on manual work of the faculty members and conserve classroom time. Educators can pay more attention to the teaching process rather than take time to mark down the attendance manually.

The system also helps students since they easily monitor records of their attendance using the system dashboard. This will assist them in monitoring their attendance status and will motivate them to attend classes.

Nonetheless, student acceptance and comfort have to be taken into account by the system too. Given that the technology of recognizing faces is being employed, the students might be first of all concerned about privacy. These concerns can be addressed by means of proper communication and understanding of how the system works.

8.2 Legal Aspects

The system processes personal data that includes details like identity of students and data of their facial images. Consequently, one should observe legal guidelines that are associated with data protection and privacy. The academic institutions should make sure that the gathered information is utilized academically.

Only authorized personnel including administrators and faculty members should have access to student data. This should have appropriate authentication systems to avoid unauthorized access.

The system should as well be in accordance with applicable data protection policies and institutional policies. Safe storage and access control would help in making sure that the system is compliant with the law.

8.3 Ethical Aspects

The ethical factor should be considered when creating systems of personal data collection. Face recognition and RFID technology should not violate student privacy and confidentiality. The students also need to know the manner in which their data is collected, stored, and utilized.

The system must not include abuses of what has been gathered or permit access to the data collected to any other end not related to attendance checking. The openness of the system operation assists in establishing trust between users.

The developers should also make sure that the system is modeled in a way that brings no discrimination to any student. The algorithm of face recognition must work effectively on the face of various people so that no one will be wrongfully treated.

8.4 Sustainability Aspects

Sustainability concentrates on long term effects of the system on both resources and environment. The proposed attendance system minimizes the use of paper based attendance registers which contributes to saving of paper in the learning institutions.

The impact on the environment is not very high since the system requires little hardware materials and utilizes primarily software. The open source software tools also make the cost of licensing lower and encourage long term sustainability.

The system will be able to be utilized over extended periods without having to regularly change hardware because of sound system design and maintenance.

8.5 Safety Aspects

The safety parameters are based on the protection of both users and system resources. Attendance system must be created in a way that will avoid loss of data and unauthorized access. Attendance records should be guarded by regular database backups.

Cams and RFID readers are hardware devices that need to be installed in an area that cannot be physically damaged. It should also have secured login to secure student sensitive data.

With proper security practices in place, the system can securely operate and maintain the information stored in the system.

Chapter 9

Conclusion

Automated Student Attendance Monitoring and Analytics System has been created to overcome the usual problem of conventional attendance tracking approaches applied in learning institutions. In most colleges, attendance is registered manually, and hence time consuming, provides room to error and even proxy attendance. In this project, it was planned to eliminate these constraints by introducing such a system that will offer face recognition and RFID authentication which will provide a more accurate and efficient method of attendance tracking. Various modules were developed and tested during system development, these include face detection module, identity verification module, RFID communication module, database management module and report generation module. The system had the capability to take student pictures using the camera and recognize faces through the image processing method. The identified faces were compared to the stored data to confirm the identity of the student. Meanwhile, RFID cards were employed as a secondary authentication measure as a guarantee that the registered attendance is valid. Upon the verification of the identity, the system was able to record the attendance with date and time and store the same in the database. The data stored was subsequently utilized to produce the attendance reports and analytics that can be obtained by the faculty and administrators. The system also provides the students with an opportunity to see their attendance status and thus enables them to be aware of their attendance percentage. The measurements made in the course of the testing revealed that the system is effective when operated in a controlled environment. It saves time used in marking attendance and minimizes the errors of a manual system. The system too offers increased transparency whereby the attendance records are digitized and can be accessed anytime they are needed. The introduction of analytics assists in highlighting poorly attended students at an early stage, which can assist in improved academic monitoring. On the whole, the project has fulfilled the central goal that is to create an efficient, reliable, and user-friendly system of attendance automatizing using modern technologies. It demonstrates the potential effectiveness of integrating software and hardware solutions into the educational system to enhance their efficiency and accuracy. The system can be effectively applied to college environments in the real world with further enhancements and adequate implementation and can help to better administer academic management.

References

- [1] A. Kumar and B. Singh, Face Recognition Based Attendance System in Proc. IEEE Int. Conf. Advanced Computing, 2020.
- [2] S. Sharma, R. Gupta, and P. Verma, "RFID Based attendance monitoring system, 2009, International Journal of Engineering Research, vol. 8, no.3, pp. 45–49, 2019.
- [3] P. Viola and M. Jones, Rapid Object Detection using a Boosted Cascade of Simple Features, in \proceedings. IEEE Conf. Computer Vision and Pattern Recognition CVPR, 2001.
- [4] T. Ahonen, A. Hadid, and M. Pietikainen, "Face Description with Local Binary Patterns," IEEE Trans. Pattern Anal. Mach. Intell., vol. 28, no. 12, pp. 2037-2041, 2006.
- [5] OpenCV Documentation, "Open Source Computer Vision Library.
- [6] N. Dalal and B. Triggs, "Histograms of Oriented Gradients for Human Detection," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2005.
- [7] NXP Semiconductors, "MFRC522 RFID Reader Datasheet," 2016.
- [8] R. Szeliski, Computer Vision: Algorithms and Applications, Springer, 2011.
- [9] A. Jain, A. Ross, and S. Prabhakar, "An Introduction to Biometric Recognition," IEEE Trans. Circuits and Systems, vol. 14, no. 1, pp. 4-20, 2004.
- [10] M. Turk and A. Pentland, Eigenfaces to Recognition, Journal of Cognitive Neuroscience, vol. 3, no. 1 pp. 71–86, 1991.
- [11] K. Delac and M. Grgic, A Survey of Biometric Recognition Methods, in Proc. IEEE Int. Symp. Electronics, 2004.
- [12] J. Smith, "Automated Attendance system with Face recognition feature,International Journal of Computer Applications, vol. 182, no. 10, p. 1–5,2021
- [13] S. Li and W. Deng, "Deep Facial Recognition: A Survey," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 44, no. 12, pp. 1–24, 2022.
- [14] Y. Taigman, M. Yang, M. Ranzato, and L. Wolf, "DeepFace: Closing the Gap to Human-Level Performance in Face Verification," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2014.

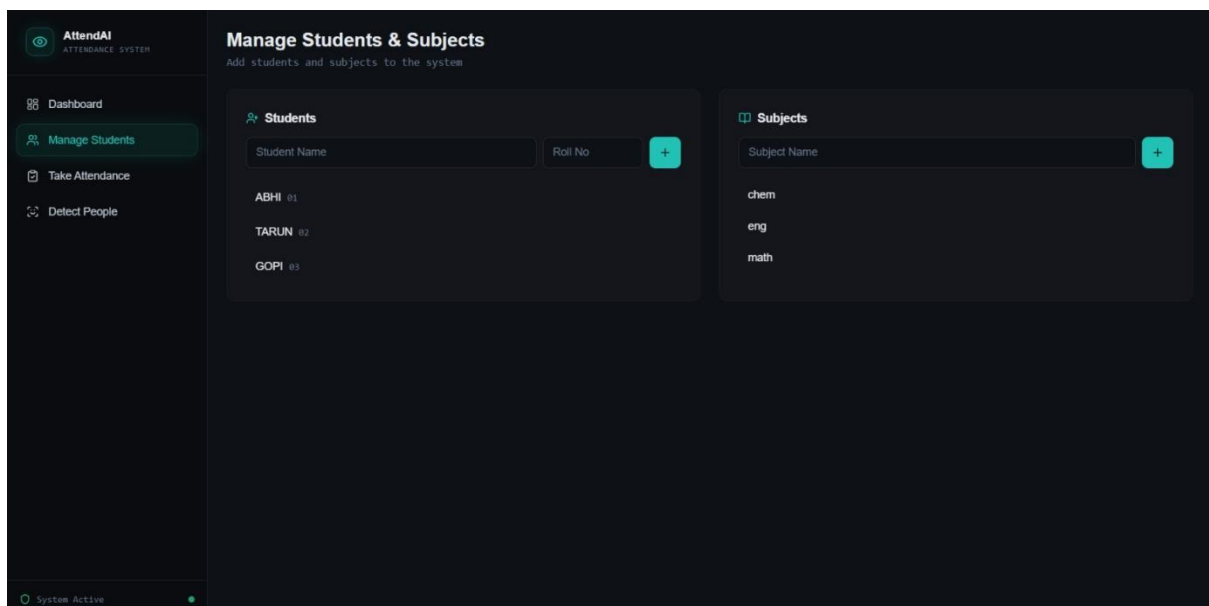
- [15] F. Schroff, D. Kalenichenko, and J. Philbin, “FaceNet: A Unified Embedding for Face Recognition and Clustering,” in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2015.
- [16] I. Goodfellow et al., “Challenges in Representation Learning: A Report on Three Machine Learning Contests,” in Neural Networks, vol. 64, pp. 59–63, 2015.
- [17] H. Wang and G. Doretto, “Student Attendance System Using Face Recognition,” International Journal of Computer Science and Information Security, vol. 17, no. 5, pp. 120–125, 2019.
- [18] R. Patel and M. Shah, “Automated Attendance System Using Face Detection and Recognition,” International Journal of Engineering Research and Technology, vol. 9, no. 6, pp. 102–106, 2020.
- [19] S. K. Singh and A. K. Tripathi, “Smart Attendance System Using Face Recognition and IoT,” International Journal of Innovative Technology and Exploring Engineering, vol. 9, no. 2, pp. 2345–2350, 2019.
- [20] P. Gupta and K. Kumar, “IoT Based Smart Attendance Monitoring System Using RFID and Face Recognition,” in Proc. IEEE Int. Conf. Smart Systems and Inventive Technology, 2021.

Appendix

Appendix A: Sample System Outputs

The following outputs were observed during the execution of the Automated Student Attendance Monitoring and Analytics System. When the attendance session is started, the system activates the camera and detects student faces in real time. Once a face is detected, it is verified with the registered dataset. If the identity matches, the system records attendance along with date and time. The output is displayed on the screen, showing successful attendance marking.

In addition to face recognition, RFID authentication was also tested. When a student scans the RFID card, the system reads the unique card ID and verifies it with the stored database. If the card is valid, the system confirms the identity and records attendance successfully



Appendix B: Database Design

The system uses a structured database to store student information and attendance records. The database consists of tables that manage student details, RFID information, and attendance data.

A]Student Table

Field Name	Type	Description
StudentID	Integer	Unique student identifier
Name	Varchar	Name of the student
RollNo	Varchar	Student roll number
RFID	Varchar	RFID card number

B]Attendance Table

Field Name	Type	Description
AttendanceID	Integer	Unique attendance record ID
StudentID	Integer	Reference to student
Date	Date	Attendance date
Time	Time	Attendance time
Status	Varchar	Present or Absent

Appendix C: Sample Test Cases

The system was tested using different scenarios to verify its performance and accuracy.

Test Case	Input	Expected Output	Result
Face Detection	Student face in camera	Face detected	Successful
Face Recognition	Registered student	Identity verified	Successful
RFID Scan	Valid RFID card	Card ID detected	Successful
Attendance Recording	Verified student	Attendance stored	Successful

Appendix D: Hardware Setup

The hardware setup of the system includes a camera, RFID reader, Arduino microcontroller, and a computer system. The camera is positioned to capture student faces clearly during attendance sessions. The RFID reader is connected to the Arduino, which reads the card data and sends it to the computer system.

The Arduino communicates with the system using serial communication. The computer processes the data received from both the camera and RFID module to verify student identity and record attendance. The setup ensures smooth interaction between hardware and software components for accurate attendance monitoring.